Heritable extrinsic frailty biases intrinsic lifespan heritability estimates in calibrate-then-extrapolate models

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Abstract. Shenhar et al. (2026) estimate that the heritability of human lifespan rises to approximately 50% after correcting for extrinsic mortality, using parametric mortality models calibrated to Scandinavian twin data. Hamilton (2026) argues that infection mortality is heritable and that concordant-survivor restriction inflates Falconer h^2; his biological argument is well-supported, but his simulation implements a different analytical procedure than Shenhar's.

Shenhar's framework may be mathematically valid under its assumptions, but its "intrinsic" estimate is not empirically identified as purely intrinsic if heritable extrinsic frailty exists and is omitted at calibration. We identify a distinct vulnerability in their calibrate-then-extrapolate framework. If individuals vary in genetic susceptibility to extrinsic mortality and this heterogeneity is not modeled, the single-parameter calibration absorbs extrinsic genetic variance into the intrinsic parameter — a form of omitted-variable bias operating at the estimation step rather than through sample restriction. In simulations matched to Danish twin cohorts (50,000 pairs per zygosity, three mortality model families), a canonical single-seed decomposition shows 11 pp of upward inflation relative to a correctly specified baseline, partially offset by 4 pp of downward attenuation from extrinsic noise (net bias +7.6 pp in that seed). Under 20-seed Monte Carlo replication, the net bias is precisely estimated and model-general (mean ± SE): Gompertz–Makeham $+9.2 \pm 0.3$ pp, Makeham-Gamma-Gompertz $+10.4 \pm 0.5$ pp, and Saturating-Removal $+9.2 \pm 0.5$ pp (all 95% CIs entirely above +8 pp); the canonical single-seed GM estimate of +7.6 pp falls within the range of individual seed variation. The bias persists without survival conditioning (+3.1 pp at cutoff = 0), is robust to the functional form of extrinsic frailty (log-normal, additive, gamma constructions yield overlapping CIs), and scales monotonically with the heritable fraction of extrinsic mortality. Four negative controls establish that the inflation requires genetic concordance in a mortality-relevant pathway with non-negligible extrinsic hazard; a positive control (correctly specified two-component model) recovers σ_θ to within −3.2% of its true value — eliminating the inflation component. Within an externally informed sensitivity regime restricted below empirically motivated levels, the net bias ranges from +2.8 to +18.9 pp (mean 9 pp); the anchored range is conservative, as bias continues to increase beyond it. Negative genetic correlation ($\rho < 0$) reverses the net bias sign — a falsifiable prediction confirmed in all three model families.

[1]Simulation code available at https://github.com/biostochastics/extrinsic-frailty-heritability/.

1 Introduction

1.1 The claim under dispute

Shenhar et al. (2026) report that the heritability of human lifespan rises from the commonly cited 20–25% to approximately 50% after correcting for extrinsic mortality — deaths from infections, accidents, and environmental hazards. Their method fits parametric mortality models to historical twin cohorts, calibrates genetic variance to reproduce observed twin correlations, then extrapolates to a counterfactual world where the extrinsic hazard m_{ex} is set to zero. The implied correction — from observed $h^2 \approx 0.20$ to intrinsic $h^2 \approx 0.50$ — is large, and any systematic bias in the calibration step propagates directly into the headline estimate.

1.2 Two distinct threats

Hamilton (2026) critiques this claim, arguing that infection mortality is heritable and that removing extrinsic mortality introduces selection bias. His simulation demonstrates that conditioning on survival from genetically-patterned extrinsic death inflates Falconer h^2. The biological argument — that susceptibility to severe infection is substantially heritable — is well-supported by the Sørensen et al. (1988) adoption study (RR = 5.81, 95% CI: 2.47–13.7, for infection death) and Danish twin studies estimating 40% heritability of infectious disease mortality (Obel et al., 2010).

However, Hamilton's simulation does not replicate Shenhar's procedure. It removes twin pairs where either member died extrinsically (concordant-survivor conditioning), whereas Shenhar does not apply concordant-survivor restriction — they parametrically set $m_{\text{ex}} = 0$ in the fitted model. These are different operations and can induce bias through distinct mechanisms (which may coexist):

- **Collider bias** (Hamilton): conditioning on survival creates a genetically non-representative sample.
- **Omitted-variable bias** (this paper): a misspecified model absorbs extrinsic genetic variance into the intrinsic parameter at the calibration step.

1.3 Our contribution

Building on Hamilton's biological insight, we identify the bias mechanism specific to Shenhar's parametric procedure and provide eight lines of evidence characterizing it: a two-component decomposition of the bias, direct observation of variance absorption, negative controls and a positive control (oracle fix) establishing specificity, demonstration that survival conditioning is not required, dose-response to extrinsic mortality, replication across all three of Shenhar's mortality models, a falsifiable sign-reversal prediction, and quantification within an externally informed sensitivity regime restricted below empirically motivated levels.

2 Conceptual framework

2.1 Shenhar's estimand and procedure

Shenhar et al. model individual mortality as:

$$\mu_i(t) = m_{\text{ex}} + h(t, \theta_i) \tag{1}$$

where m_{ex} is a cohort-level constant (identical for all individuals) and θ_i are individual intrinsic aging parameters drawn from a Gaussian distribution with standard deviation σ_θ. For MZ twins, both members share the same θ; for DZ twins, their θ values are correlated at $r = 0.5$.

The procedure has two steps:

1. **Calibration:** adjust σ_θ so that simulated MZ twin lifespan correlations match those observed in historical twin data (at historical m_{ex}).
2. **Extrapolation:** set $m_{\text{ex}} = 0$, keep the calibrated σ_θ, and recompute Falconer $h^2 = 2(r_{\text{MZ}} - r_{\text{DZ}})$.

Under their assumptions — that m_{ex} is truly a non-genetic population constant — this extrapolation is mathematically valid.

2.2 Omitted heritable extrinsic susceptibility

Suppose the true data-generating process is:

$$\mu_i(t) = m_{\text{ex}} \cdot \exp\big(\sigma_\gamma \gamma_i - \sigma_\gamma^2/2\big) + \exp(\theta_i) \cdot \exp(b \cdot t) \tag{2}$$

where $\gamma_i \sim \mathcal{N}(0, 1)$ is an individual's genetic susceptibility to extrinsic mortality (the centering term $-\sigma_\gamma^2/2$ ensures $\mathbb{E}\big[\exp\big(\sigma_\gamma \gamma_i - \sigma_\gamma^2/2\big)\big] = 1$, so that $\mathbb{E}[c_i] = m_{\text{ex}}$ for all σ_γ) and $\theta_i \sim \mathcal{N}\big(\mu_\theta, \sigma_\theta^2\big)$ is the intrinsic aging parameter. MZ twins share the same γ (and θ); DZ twins' values are correlated at 0.5. The within-individual correlation $\text{Corr}(\theta_i, \gamma_i) = \rho$ represents pleiotropy — shared genetic pathways between immune function and intrinsic aging.

Shenhar's model cannot represent individual extrinsic heterogeneity. When the calibration step matches σ_θ to observed r_{MZ}, it must account for twin concordance contributed by **both** shared intrinsic genetics (through θ) **and** shared extrinsic genetics (through γ). Since the model has only one source of genetic variation (σ_θ), it absorbs both. The calibrated σ_θ is inflated.

Because θ and γ enter different terms of the hazard, the relationship between fitted and true variances is model-dependent. As a first-order heuristic:

$$\sigma^2_{\theta,\text{fit}} \approx \sigma^2_{\theta,\text{true}} + \alpha\sigma_\gamma^2 + 2\beta\rho \cdot \sigma_{\theta,\text{true}} \cdot \sigma_\gamma \tag{3}$$

where α and β are model-dependent coefficients. The $\alpha = \beta = 1$ simplification provides a reasonable first-order approximation (mean absolute residual 8.6%; see §4.2 below). A formal delta-method derivation establishing conditions for inflation is given in Section B (Proposition B1).

This is fundamentally a structural identifiability problem: when the true hazard contains two heritable components but the fitted model contains only one, σ_θ becomes the model's single knob for explaining all genetic concordance in mortality. Under misspecification, any standard fitting criterion converges to a pseudo-true parameter that best approximates the joint MZ/DZ survival structure within the restricted model class. Positive pleiotropy ($\rho > 0$) ensures that the inflation is generically upward, because the omitted extrinsic component pushes twin concordance in the same direction as intrinsic frailty. The exact magnitude depends on the fitting criterion, but the mechanism is structural (see §3.3 and §4.2 for simulation evidence; Section B for a formal derivation).

2.3 Distinction from Hamilton's mechanism

Hamilton demonstrates collider bias from concordant-survivor conditioning — removing extrinsically-dead pairs creates a genetically non-representative sample. Shenhar's procedure does not do this; our bias enters at calibration and persists with cutoff = 0 (§4.4). The two mechanisms may coexist but are analytically distinct.

Table 1: Simulation parameters matching Danish cohorts born 1870–1900.

Parameter	Value
Mortality model	Gompertz-Makeham: $\mu(t) = c + a \cdot \exp(bt)$
Gompertz slope b	0.085 (fixed)
Mean μ_θ	$\log(5 \times 10^{-5})$
Historical m_{ex}	0.004 per year
Cutoff age	15 (Danish twin registry convention)
Twin pairs	50,000 per zygosity
σ_θ	1.259 (auto-calibrated for oracle $h^2 \approx 0.50$)
σ_γ	0.40 (default); exploratory sweep 0.15–0.55 by 0.05 (9 levels); anchored sweep 0.30–0.65 by 0.05 (§3.4, §4.7)
ρ (pleiotropy)	0.4 (default); exploratory sweep 0–0.8 by 0.1 (9 levels); anchored sweep 0.20–0.50 by 0.05 (§3.4, §4.7)

3 Methods

3.1 Simulation overview

We construct a simulation that replicates Shenhar's calibrate-then-extrapolate procedure under two data-generating processes: one where their assumptions hold, and one where they are violated by heritable extrinsic susceptibility. All results use 50,000 twin pairs per zygosity. Lifespans are simulated by inverting the cumulative hazard via the Lambert W function in closed form.

3.2 Arms and controls

Our experimental design follows a structured hierarchy: a **reference oracle**, **primary arms** that test the central hypothesis against its null, **negative controls** that isolate necessary conditions, a **positive control** that confirms recoverability, and **diagnostic sweeps** that probe boundary behavior. Table 2 summarizes all conditions; the subsections below describe each in detail.

3.2.1 Reference

Oracle. The oracle runs the pure intrinsic DGP ($\sigma_\gamma = 0$, $m_{\text{ex}} = 0$) using the true $\sigma_\theta = 1.259$, calibrated to yield $h^2 \approx 0.50$. No calibrate-then-extrapolate pipeline is applied — the oracle evaluates Falconer h^2 directly from intrinsic-only twin correlations, providing the ground-truth benchmark ($h^2 = 0.51$) against which all other conditions are measured.

3.2.2 Primary arms

Arm 1 — Correctly specified baseline. Data are generated under Shenhar's own assumptions: extrinsic mortality is a population-level constant ($c_i = m_{\text{ex}} = 0.004$ for all individuals), with no individual heterogeneity ($\sigma_\gamma = 0$). The model is **correctly specified**. We calibrate σ_θ to match the simulated r_{MZ} via 40-iteration stochastic bisection, then extrapolate to $m_{\text{ex}} = 0$. Any deviation from the oracle reflects the estimand mismatch between "intrinsic" (no extrinsic mortality present) and "extrapolated" (extrinsic mortality present during calibration, then removed). This arm validates that the calibrate-then-extrapolate logic is internally consistent.

Arm 2 — Misspecification (main result). Data are generated under the two-component DGP (Equation (2)): extrinsic mortality is individually heterogeneous ($\sigma_\gamma = 0.40$, $\rho = 0.4$), heritable, and assigned at the individual level. Shenhar's **one-component** model — which lacks any γ parameter — is then fit by the same calibration pipeline as Arm 1. This is the central test: does omitting heritable extrinsic susceptibility from the model inflate the intrinsic estimate?

Arm 3 — Hamilton conditioning. Hamilton's procedure removes twin pairs where either member died extrinsically before a threshold age (concordant-survivor restriction). We implement this as a separate arm for comparison: data follow the same two-component DGP as Arm 2, but instead of calibrating a parametric model, we compute Falconer h^2 directly from the surviving concordant subsample. This represents a different analytical procedure and a different bias mechanism (collider bias from sample restriction rather than variance absorption from calibration).

3.2.3 Negative controls

Control A — Non-heritable extrinsic. Data include individual extrinsic heterogeneity ($\sigma_\gamma = 0.40$), but γ values are drawn **independently** for each twin rather than shared within pairs (`gamma_heritable = FALSE`). The extrinsic component adds individual mortality variance but no **genetic** concordance. If the calibration bias requires within-pair correlation in extrinsic susceptibility, this control should show no inflation.

Irrelevant trait (ζ). A heritable latent trait ζ is genetically correlated with θ at the same level as γ in Arm 2, but ζ does not enter the hazard function — it has no effect on mortality. This tests whether shared genetic architecture **per se** is sufficient to bias calibration, or whether the correlated trait must affect mortality through the hazard.

Control B — Vanishing m_{ex}. Data include heritable extrinsic susceptibility ($\sigma_\gamma = 0.40$, $\rho = 0.4$) — the same two-component DGP as Arm 2 — but with negligible extrinsic mortality magnitude ($m_{\text{ex}} = 0.0002$, one-twentieth of historical levels). This tests the boundary condition: when the extrinsic hazard channel carries negligible mortality signal, the bias should vanish regardless of how strongly extrinsic susceptibility is heritable.

Pleiotropy isolation ($\rho = 0$). Same DGP as Arm 2 except $\rho = 0$: extrinsic susceptibility is heritable and individually heterogeneous, but genetically **independent** of intrinsic frailty. This isolates the contribution of pure extrinsic concordance from the amplifying effect of pleiotropy.

3.2.4 Positive control

Oracle fix — Correctly specified two-component model. Data are generated under the full two-component DGP (same as Arm 2), but calibration uses a model that **includes** the extrinsic frailty component: (σ_γ, ρ) are held at their true DGP values, and only σ_θ is estimated via bisection on r_{MZ}. If the bias is genuinely caused by omitting γ from the model, then including it should recover the true σ_θ and eliminate the inflation component, leaving only the attenuation that arises under correct specification. This is the critical positive control — the mirror image of Arm 2.

3.2.5 Diagnostic sweeps

Several additional analyses probe boundary behavior and dose-response (full results in §4):

- **Cutoff = 0:** Repeats Arm 2 with no age cutoff — all twin pairs included, including infant deaths. Tests whether the bias requires concordant-survivor conditioning.
- **Dose-response:** Sweeps m_{ex} from 0 to 0.012 (3× historical) with all other parameters at Arm 2 values.
- **Negative ρ sweep:** Sweeps the genetic correlation ρ from -0.60 to $+0.80$, testing the prediction that negative ρ reverses the bias sign.

Table 2: Full simulation design. All conditions use 50,000 twin pairs per zygosity. σ_γ, ρ, and heritability refer to the **data-generating process**; the fitted model is always Shenhar's one-component specification except for the oracle fix. Arm 2 row is highlighted.

Condition	σ_γ	ρ	m_{ex}	Heritable?	Tests
Oracle	0	—	0	—	Ground-truth intrinsic h^2
Arm 1: Correctly specified	0	—	0.004	—	Pipeline consistency under correct model
Arm 2: Misspecification	0.40	0.4	0.004	Yes	Central hypothesis — variance absorption
Arm 3: Hamilton conditioning	0.40	0.4	0.004	Yes	Collider bias comparison (different procedure)
Control A: Non-heritable	0.40	0	0.004	No	Specificity — requires genetic concordance?
Irrelevant trait (ζ)	—	$\rho_\zeta = 0.4$	0.004	Yes	Specificity — trait must enter hazard?
Control B: Vanishing m_{ex}	0.40	0.4	0.0002	Yes	Boundary — requires extrinsic signal?
Pleiotropy isolation	0.40	0	0.004	Yes	Isolates $\rho = 0$ from $\rho > 0$ effect
Oracle fix: correct model	0.40	0.4	0.004	Yes	Positive control — model recovers σ_θ?
Cutoff = 0	0.40	0.4	0.004	Yes	Boundary — requires survival conditioning?

- **m_{ex} split:** Decomposes m_{ex} into heritable and non-heritable fractions, sweeping the heritable share from 0 to 100%.

3.3 Calibration and estimands

The primary calibration uses stochastic bisection on Monte Carlo estimates of r_{MZ} (40 iterations). This matches Shenhar et al.'s procedure: their supplement explicitly states that "parameter heterogeneity was adjusted to match MZ twin lifespan correlations by varying the degree of parameter variation" — confirming single-target r_{MZ} calibration. They also report Falconer $h^2 = 2(r_{\text{MZ}} - r_{\text{DZ}})$, making our summary statistic directly comparable to their published estimates.

A natural concern is whether calibrating to r_{MZ} alone understates the information available — perhaps a joint $(r_{\text{MZ}}, r_{\text{DZ}})$ calibration would behave differently. We tested this directly: under the misspecified DGP, the model fitted to r_{MZ} also predicts r_{DZ} with near-zero discrepancy (0.0002). A joint calibration targeting both correlations yields $\sigma_{\text{fit}} = 1.552$ and bias = +9.3 pp — slightly larger than the r_{MZ}-only result ($\sigma_{\text{fit}} = 1.537$, bias = +7.6 pp), consistent with the additional r_{DZ} target pulling σ_θ further upward to accommodate genetic concordance from both zygosity groups simultaneously. This shows that adding r_{DZ} information does not resolve the absorption — and if anything slightly amplifies it: the misspecified model's single genetic parameter must account for all genetic concordance regardless of whether one or two correlation targets are used. Full-likelihood fitting can improve diagnostics (e.g., posterior predictive checks on bivariate survival surfaces), but it cannot recover a decomposition that the model does not parameterize. Under an underparameterized one-component model, maximum

likelihood will still select the parameter value that best approximates the true two-component data within the restricted model class — i.e., a pseudo-true σ_θ that absorbs part of the omitted heritable extrinsic component. Indeed, Wienke et al. (2003) apply maximum-likelihood correlated gamma-frailty models to the same Danish twin cohorts and still estimate large familial frailty contributions (frailty heritability ≈ 0.58), confirming that the phenomenon is not an artifact of our specific calibration criterion.

3.4 Anchoring σ_γ and ρ to independent evidence

Anchor for σ_γ. Obel et al. (2010) report liability-scale heritability $h_L^2 \approx 0.40$ (95% CI: 0.12–0.50) for infection mortality in 44,005 Danish twin pairs. This is consistent with the substantial familial aggregation reported by Sørensen et al. (1988 RR = 5.81 for adoptee infection death) and extended by Petersen et al. (2010), who show genetic influences on both infection incidence and case fatality in Danish adoptees. As an order-of-magnitude anchor, $\sigma_\gamma \approx \sqrt{h_L^2} \approx 0.63$ at $h_L^2 = 0.40$. This mapping is approximate — liability-scale and log-hazard dispersion are not on identical scales — and we evaluate robustness to this translation by sweeping $\sigma_\gamma \in \{0.30, 0.40, 0.55, 0.63\}$.

Corroborating evidence comes from correlated frailty models fit directly to Scandinavian twin survival data: Wienke et al. and Yashin et al. estimate frailty heritabilities of ≈ 0.58 (Danish twins) and comparable magnitudes (Swedish twins), respectively (see also §3.3). These estimates reflect total unobserved frailty, not an extrinsic channel per se, but they confirm that substantial familial heterogeneity in mortality susceptibility exists in the same cohort context — supporting the empirical plausibility of our sensitivity range.

Anchor for ρ. Qiu (2025) report genome-wide genetic correlations between COVID-19 severity and longevity of $r_g \approx -0.34$ to -0.35, corroborated by Mendelian randomization (Ying et al., 2021 genetically-proxied 10-year lifespan increase reduces COVID-19 infection risk, OR = 0.31). The shared architecture is driven by conserved inflammatory pathways (IL-6/IL-6R, TNF-α, NF-κB, FOXO3, APOE, TLR) with documented pleiotropic effects on immune function and cellular senescence (Franceschi et al., 2000; Rosa et al., 2019). Because our θ parameterizes log-hazard (higher θ increases mortality, decreasing longevity), a negative genetic correlation between **longevity** and infection severity corresponds to a **positive** correlation between the two mortality-hazard components — i.e., $\mathrm{Corr}(\mathrm{severity}, \theta) = -\mathrm{Corr}(\mathrm{severity}, \mathrm{longevity})$, hence $\rho > 0$ in our notation. We adopt $\rho \in [0.20, 0.50]$. COVID-19 represents a single novel pathogen and serves as an existence proof that genetic correlation between infection susceptibility and aging pathways exists — not as a magnitude anchor for the pre-antibiotic Danish cohorts, whose mortality was dominated by tuberculosis, pneumonia, diphtheria, and enteric infections with different immunological mechanisms. Stronger evidence comes from GWAS: Schurz et al. (2024) estimate the SNP-heritability of TB susceptibility at 26.3% (95% CI 23.7–29.0%) with the only genome-wide significant signal in the HLA class II region (HLA-DQA1/DRB1). Independently, Joshi et al. (2017) identify HLA-DQA1/DRB1 as genome-wide significant for parental lifespan — a proxy longevity phenotype — in 606,059 individuals. The convergence of signals in the same HLA class II region across the two traits is consistent with shared immunogenetic architecture linking infection susceptibility to aging-related mortality, though formal colocalization has not been performed. In Nordic populations specifically, HLA-DQB1*05:01 is protective for TB (OR = 0.81; (Tervi et al., 2023)), and the Sørensen adoption study found RR = 5.81 (95% CI: 2.47–13.7) for infection death among biological relatives — concentrated in case-fatality rather than incidence, with evidence for non-additive genetic effects (Petersen et al., 2010; Sørensen et al., 1988). The shared inflammatory architecture (IL-6/IL-6R, TNF-α, NF-κB, FOXO3, APOE, TLR) connects immune function to cellular senescence across pathogen classes (Franceschi et al., 2000; Rosa et al., 2019). No direct LD-score regres-

sion estimate between pre-antibiotic infection mortality and longevity exists; our ρ range is informed by the convergence of these indirect lines of evidence, not calibrated to any single estimate. The true genetic correlation between **general** historical extrinsic susceptibility and intrinsic aging may differ from the COVID-19-specific estimate in either direction.

Bridging σ_γ to liability-scale h^2_L. The mapping $\sigma_\gamma \approx \sqrt{h^2_L}$ is heuristic — liability-scale heritability depends on prevalence and the threshold model, while σ_γ parameterizes a continuous log-hazard frailty. To ground the translation, we run the full competing-risks DGP at each value of σ_γ, generate MZ and DZ twin pairs, create a binary indicator for whether each twin died of infection, compute tetrachoric correlations from the resulting 2×2 tables, and derive Falconer's h^2_L (Figure 7). Under our assumed hazard structure, matching Obel et al.'s $h^2_L = 0.40$ requires $\sigma_\gamma \approx 1.47$ — more than double our upper bound of 0.65. We treat this bridge as directional evidence — confirming that published infection-mortality heritability implies substantial extrinsic heterogeneity, and that our sensitivity range sits well below the empirically implied parameterization — but not as a calibration target for σ_γ. The bridge inherits the DGP's simplifications (single competing risk, stylized hazard structure, liability-threshold mapping), so the specific σ_γ value it implies should not be taken at face value. What it does establish is the sign and relevance of the effect: extrinsic heterogeneity of the magnitude implied by published epidemiological data would produce biases at least as large as those we report.

Externally informed restricted sensitivity regime: $\sigma_\gamma \in [0.30, 0.65]$, $\rho \in [0.20, 0.50]$. We cap σ_γ at 0.65 — well below the bridge-implied value of ≈ 1.47 — so that the reported biases reflect what the stylized DGP produces under cautious assumptions rather than extrapolations from a bridge we treat as directional only. An extended stress test (§4.7, Figure 7 panel B) confirms that bias continues to increase monotonically beyond this ceiling; the formal proof's equal-noise approximation (Proposition B1) also becomes quantitatively unreliable in that regime. The correlated-frailty literature (Wienke et al., 2003; Yashin et al., 1999) independently supports the existence of substantial familial heterogeneity in these cohorts, reinforcing the qualitative conclusion that the sensitivity range sits below empirically motivated levels.

4 Results

The misspecification bias has a rich internal structure. We present eight lines of evidence characterizing its components, boundary conditions, and magnitude, and report replication-based robustness and uncertainty analyses throughout.

4.1 Two-component bias decomposition

The bias decomposes into two competing forces operating in opposite directions. Under correct specification (Arm 1: no extrinsic heterogeneity), the calibrate-then-extrapolate procedure slightly **underestimates** intrinsic heritability: observed $h^2 = 0.472$ versus oracle $h^2 = 0.51$, a gap of −3.7 pp (Figure 1). This attenuation reflects the fact that extrinsic mortality compresses twin correlations even when the model is correctly specified — an estimand shift, not a pipeline error. Arm 1 shows the calibrate-then-extrapolate logic is internally consistent under correct specification.

Under misspecification (Arm 2: heritable extrinsic susceptibility with $\sigma_\gamma = 0.40$, $\rho = 0.4$), the calibrated σ_θ inflates by 22.1% — from 1.2588 to 1.537 — producing $h^2 = 0.586$, an inflation of +11.4 pp relative to the Arm 1 baseline. This more than offsets the attenuation, yielding a net upward bias of +7.6 pp. Throughout, we refer to the downward extrinsic-noise term as the **attenuation component** and the

upward variance-absorption term as the **inflation component**; their sum is the **net bias**. The 11.4 pp inflation component is the methodological vulnerability.

Notation. This section develops the two-component decomposition using a single canonical seed (MASTER_SEED = 66829) under the Gompertz–Makeham model; the same canonical seed is used for all controls and diagnostic analyses throughout, where seed-specific structure is informative. Replicated means with Monte Carlo confidence intervals — reported in §4.6 — are used for uncertainty quantification and cross-model generality claims.

4.2 Variance absorption diagnostic

To confirm that the inflation reflects genuine variance absorption — rather than some other artifact — we compare the calibrated $\sigma^2_{\theta,\text{fit}}$ against the prediction from the omitted-variable heuristic (Equation (3)) across all 81 cells of the $\rho \times \sigma_\gamma$ sensitivity grid. If the mechanism is variance absorption, then the fitted parameter should track the predicted value $\sigma^2_{\theta,\text{true}} + \sigma^2_\gamma + 2\rho\sigma_{\theta,\text{true}}\sigma_\gamma$.

It does. The $\alpha = \beta = 1$ approximation yields a mean absolute residual of 8.6% (Figure 2). An OLS fit estimates $\alpha = 1.66$, $\beta = 1.25$ with $R^2 = 0.988$, confirming that the calibrated parameter is not arbitrarily wrong — it converges to a pseudo-true value that systematically incorporates both the omitted extrinsic frailty variance and its covariance with intrinsic frailty. The empirical coefficients exceed unity ($\alpha > 1$, $\beta > 1$) because the Gompertz hazard structure amplifies the effective contribution of each frailty source: both θ and γ enter through exponential terms ($\exp(\theta)$ and $\exp(\sigma_\gamma\gamma)$), so their interaction in the cumulative hazard produces larger-than-linear effects on twin concordance. The heuristic (Equation (3)) assumes additive variance contributions, but the multiplicative hazard structure creates a convexity that inflates the empirical coefficients above the $\alpha = \beta = 1$ approximation.

4.3 Specificity: negative controls

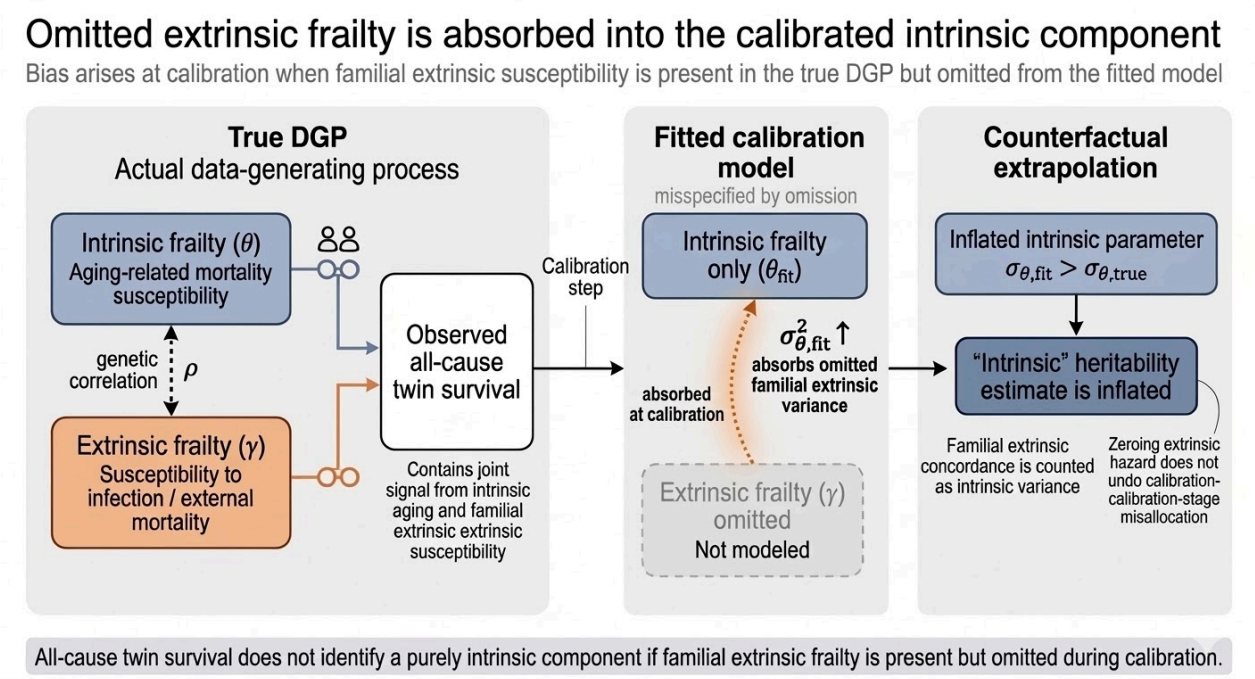
In our DGP, inflation arises from **heritable** extrinsic susceptibility that affects mortality; pleiotropy ($\rho > 0$) amplifies the effect, but is not required. Mere additional latent genetic structure is not sufficient. Four negative controls establish this.

When extrinsic susceptibility exists but is independent between twins (Control A: γ not shared), $h^2 = 0.478$ (–3.1 pp), matching the Arm 1 baseline: non-genetic noise adds individual variance but not twin concordance. When a heritable trait ζ is genetically correlated with θ at the same level as γ in Arm 2 but **does not enter the hazard function**, $h^2 = 0.502$ (–0.7 pp, within Monte Carlo noise): the mechanism requires a mortality-relevant pathway. And when heritable γ is present but m_{ex} is negligible (Control B: $m_{\text{ex}} = 0.0002$, one-twentieth of historical levels), $h^2 = 0.5$ (–0.9 pp): the extrinsic channel must carry non-negligible signal.

The inflation component is therefore specific: it requires (a) genetic concordance in extrinsic susceptibility,[2] (b) a mortality-relevant pathway, and (c) sufficient extrinsic hazard magnitude. Removing any one of these conditions eliminates the inflation. In the empirically relevant regime ($\sigma_\gamma = 0.40$), pleiotropy ($\rho > 0$) amplifies the effect substantially: heritable extrinsic susceptibility without pleiotropy ($\rho = 0$) changes the calibrated σ_θ by only 9.3%. Notably, even at $\rho = 0$ the net bias is still positive (+1.9 pp): heritable extrinsic frailty alone — without any shared genetic architecture with intrinsic aging — produces a small but non-zero inflation because the calibration absorbs extrinsic concordance that the model cannot represent.

[2]The mechanism generalizes to any source of familial concordance in extrinsic susceptibility, including shared environment. The Falconer decomposition separates genetic from shared-environmental effects through MZ/DZ differences, but the calibration step conflates any within-pair extrinsic concordance regardless of its source.

(A)



(B)

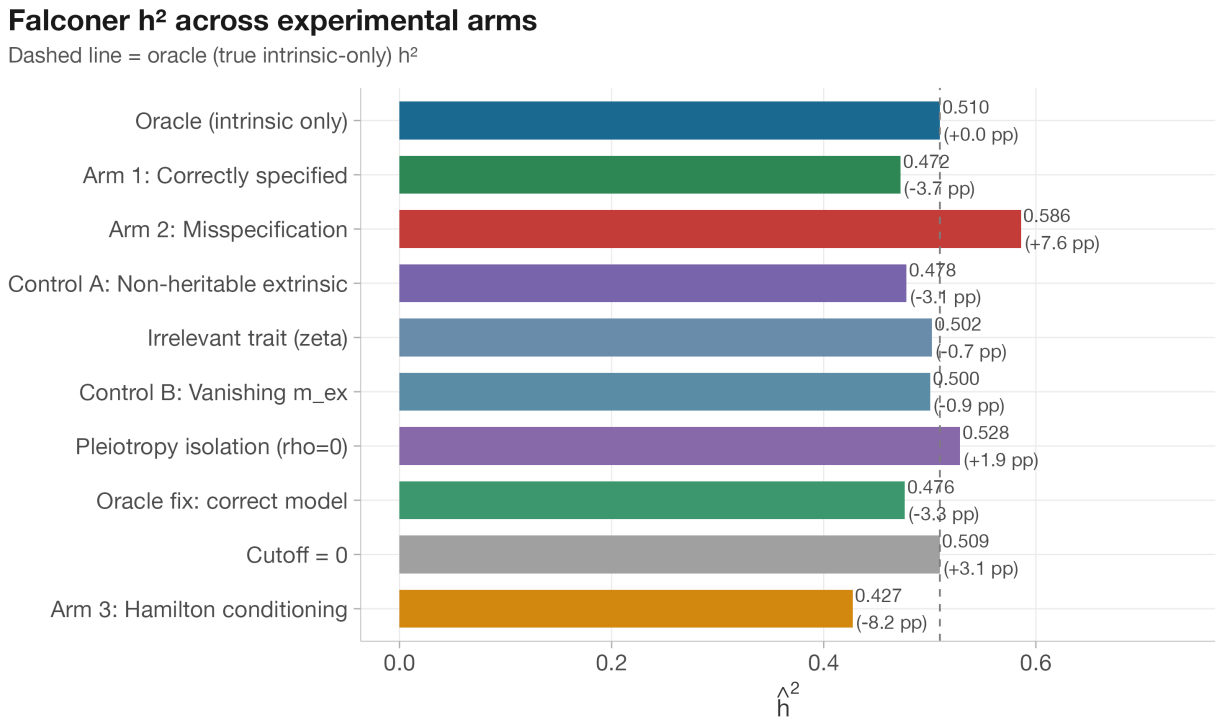


Figure 1: Mechanism and canonical worked example. **(A)** Schematic of the bias mechanism: when heritable extrinsic frailty (γ) exists in the true DGP but is omitted from the fitted model, the calibration step absorbs familial extrinsic concordance into the intrinsic parameter σ_θ, inflating the extrapolated "intrinsic" heritability estimate. **(B)** Falconer h^2 across all simulation conditions (Gompertz–Makeham, single canonical seed). Dashed line: oracle intrinsic $h^2 = 0.51$. Arm 2 (misspecified) shows net upward bias of +7.6 pp. Correctly specified conditions (Arm 1, Oracle fix) cluster near the oracle minus the 4 pp estimand-shift attenuation. Negative controls confirm specificity (§4.3); cutoff = 0 shows bias persists without survival conditioning.

Table 3: Summary of all simulation results (Gompertz–Makeham). Arm 2 (highlighted) shows net upward bias from model misspecification. Negative controls (A, irrelevant trait, B, pleiotropy isolation) confirm specificity. Oracle fix confirms recoverability. Cutoff = 0 shows the net bias persists without survival conditioning. Arm 3 shows Hamilton's collider-bias mechanism for comparison.

Condition	$\sigma_{\theta,\text{fit}}$	h^2	Bias (pp)
Oracle (intrinsic only)	1.259	0.51	—
Arm 1: Correctly specified	1.22	0.472	−3.7
Arm 2: Misspecification ($\rho = 0.4$, $\sigma_\gamma = 0.40$)	**1.537**	**0.586**	**+7.6**
Control A: Non-heritable extrinsic	1.259	0.478	−3.1
Irrelevant trait (ζ)	1.264	0.502	−0.7
Control B: Vanishing m_{ex} (0.0002)	1.267	0.5	−0.9
Pleiotropy isolation ($\rho = 0$)	1.376	0.528	+1.9
Oracle fix: correct model	1.219	0.476	−3.3
Cutoff = 0 (no age restriction)	1.323	0.509	+3.1
Arm 3: Hamilton conditioning	N/A	0.427	−8.2

A positive control complements these negative controls. If the net bias is genuinely caused by omitting γ from the model, then a **correctly specified** model — one that includes extrinsic frailty with its true parameters — should recover the true σ_θ under the same calibration pipeline. We test this by holding (σ_γ, ρ) at their true DGP values and calibrating only σ_θ to match the observed r_{MZ} from the misspecified data. The recovered $\sigma_\theta = 1.219$ (−3.2% from the true value), yielding $h^2 = 0.476$ (−3.3 pp bias). The residual bias matches Arm 1's attenuation (−3.7 pp): both are correctly specified models subject to the same estimand shift from extrinsic noise. Crucially, the **inflation component** is eliminated — σ_θ is recovered to within −3.2% of its true value. This confirms that the net upward bias in Arm 2 is specifically attributable to the omitted extrinsic frailty component and is resolvable by correctly specifying the model.

4.4 Bias without concordant-survivor restriction

A critical distinction from Hamilton's collider bias is that our mechanism does not require removing any individuals from the analysis. When we repeat the full Arm 2 analysis with cutoff = 0 — all twin pairs included regardless of age at death, including infant deaths — the misspecification bias is +3.1 pp, reduced relative to +7.6 pp with the standard age-15 cutoff, but still clearly positive. The additional bias from the cutoff reflects a genuine amplification: pairs where both twins survive past 15 are genetically non-representative when extrinsic mortality is heritable. But concordant-survivor restriction is not **necessary** for the effect. The variance absorption mechanism operates at the calibration step regardless of whether the sample is restricted — the key distinction from Hamilton's mechanism, which requires literal removal of individuals.

4.5 Dose-response to extrinsic mortality

Across our sweep, bias increases with the magnitude of extrinsic mortality (Figure 3). It is indistinguishable from zero below $m_{\text{ex}} \approx 0.001$, rises to +6.4 pp at historical levels ($m_{\text{ex}} = 0.004$; slightly below the main Arm 2 estimate of +7.6 pp because the sweep uses an independent simulation draw), and

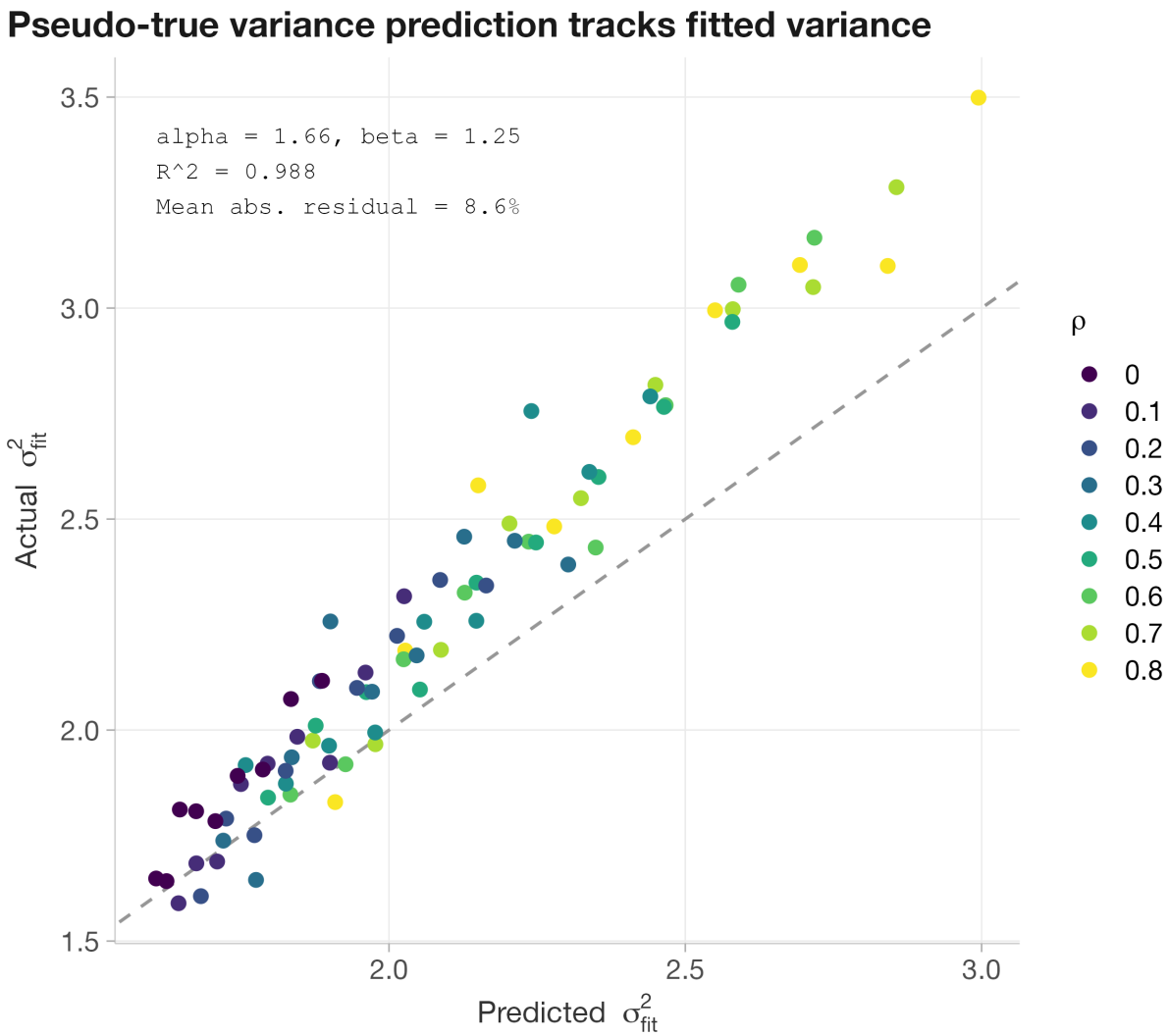


Figure 2: Pseudo-true variance prediction tracks fitted variance. Each point is one (ρ, σ_γ) cell. Points cluster near the identity line (mean absolute residual = 8.6%). Empirical coefficients: $\alpha = 1.66$, $\beta = 1.25$, $R^2 = 0.988$.

reaches +28.8 pp at $m_{\text{ex}} = 0.012$ (3× historical). This is the expected dose-response: extrinsic hazard must contribute materially to the total hazard over adult ages for genetically-patterned susceptibility to generate appreciable twin concordance. It also implies an empirical prediction: **the net bias should be larger in cohorts where infection mortality constitutes a larger share of total deaths**.

A natural objection is that not all extrinsic mortality is heritable — accidents, war, and famine are largely non-genetic. We test this by decomposing $m_{\text{ex}} = m_{\text{inf}} + m_{\text{other}}$, where only the infectious fraction m_{inf} receives genetic modulation while m_{other} remains a population constant. With replication (5 independent seeds per fraction level), Figure 4 shows that bias increases monotonically with the heritable share, crossing zero near 15–20% heritable fraction: at 0% heritable, bias is −0.7 pp (indistinguishable from zero); at 50%, 1.1 pp; and at 100%, 6 pp. The monotonic relationship implies that if even a moderate minority of historical extrinsic deaths were influenced by heritable frailty, the distortion is positive. Even if only half of historical extrinsic mortality was heritable, the mechanism produces quantitatively meaningful bias (Appendix, Section A.9).

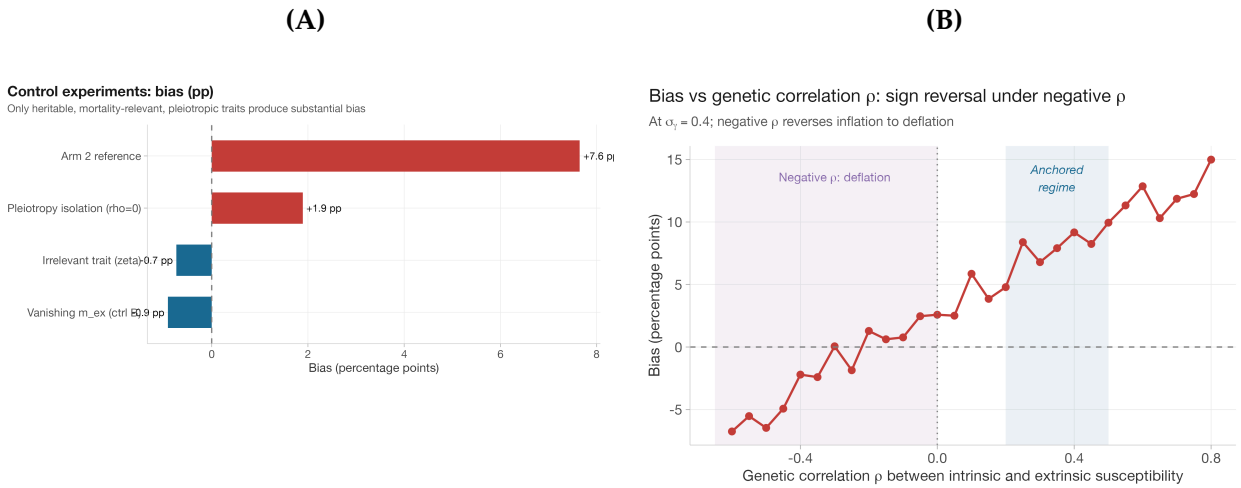


Figure 3: Specificity and sign reversal. **(A)** Negative control experiments: net bias (pp) for each control condition; Arm 2 reference: +7.6 pp. Only heritable, mortality-relevant traits with non-negligible extrinsic hazard produce substantial inflation. **(B)** Bias vs genetic correlation ρ at $\sigma_\gamma = 0.40$. Negative ρ reverses the net bias sign — a falsifiable prediction. The monotone relationship confirms that the direction of the genetic correlation determines whether the omitted extrinsic frailty inflates or deflates the intrinsic estimate.

4.6 Model and distributional generality

The mechanism is not specific to the Gompertz-Makeham functional form. We reimplemented Shenhar's two other mortality models — the Saturating-Removal (SR) model and the Makeham-Gamma-Gompertz (MGG) model — using their published parameters, and ran Arm 1 and Arm 2 under each. The SR model simulates a stochastic damage trajectory $dx = (\eta t - \beta x/(x+\kappa))dt + \sqrt{2\varepsilon}dW$, with death at a genetically variable threshold $X_c \sim \mathcal{N}(17, 3.57^2)$ and extrinsic mortality as an independent competing risk (Euler-Maruyama at weekly resolution; 50,000 pairs per zygosity). The MGG model uses Strehler-Mildvan coupling: $b_i = q_i b_0,\ \log a_i = q_i \log a_0\ \ (a_0 = 10^{-5},\ b_0 = 0.115,\ q \sim \mathcal{N}(1, 0.27^2))$. Under the correctly specified null, both models approximately reproduce Shenhar's published twin

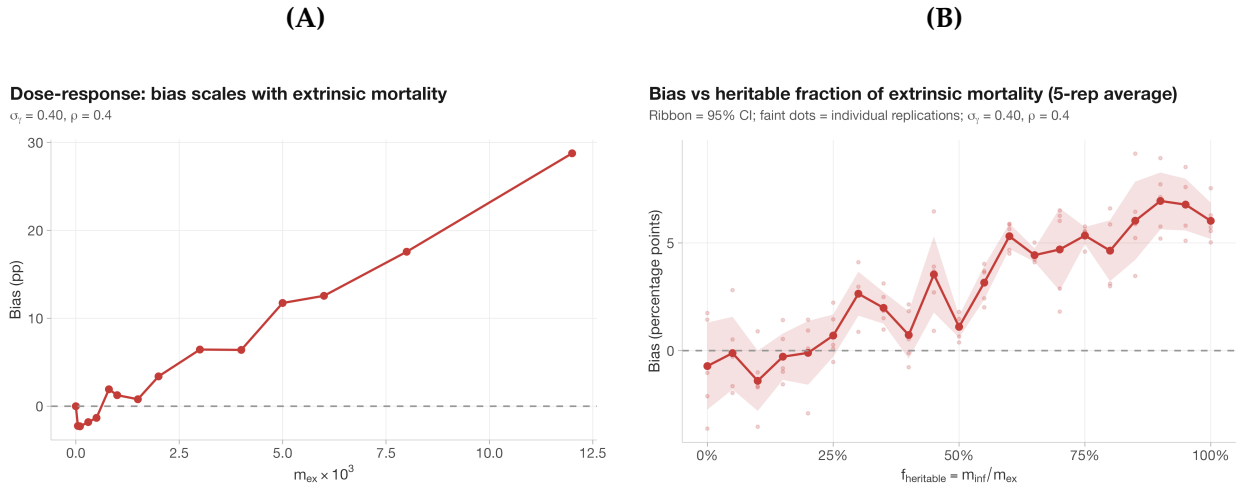


Figure 4: Dose-response structure. **(A)** Bias vs extrinsic mortality magnitude m_{ex}: bias rises from negligible at $m_{\text{ex}} < 0.001$ to +28.8 pp at 3× historical levels. **(B)** Bias vs heritable fraction of extrinsic mortality ($m_{\text{ex}} = 0.004$; 5 replications per level; ribbon = 95% CI; faint dots = individual seeds). Bias increases monotonically with the heritable share, crossing zero near 15–20%. Together, the two panels show that bias requires both material extrinsic hazard magnitude and a heritable component.

correlations and heritability estimates (Arm 1 bias < 1 pp for all three).[3]

To rigorously quantify Monte Carlo uncertainty, we replicated each model arm across 20 independent seeds (Figure 5; Appendix, Section A.10). Under correct specification (Arm 1), bias is indistinguishable from zero across all model families: GM -0.1 pp, MGG $+0.9$ pp, SR $+0.2$ pp. (Each model's replication is centered on its own oracle, so the near-zero Arm 1 means are consistent with the canonical single-seed GM attenuation of -3.7 pp reported in §4.1, which reflects the specific Gompertz–Makeham decomposition rather than a replicated aggregate.) Under misspecification (Arm 2), the replicated means (mean $\pm$ SE) are entirely positive and mutually overlapping: GM $+9.2 \pm 0.3$ pp, MGG $+10.4 \pm 0.5$ pp, SR $+9.2 \pm 0.5$ pp, with all 95% CIs (± 1.96 SE) above $+8$ pp. The vulnerability is structural: none of the three models attach an individual subscript to m_{ex}. Late-age mortality deceleration (SR's saturating removal, MGG's demographic selection) changes **how much** extrinsic frailty is absorbed but cannot eliminate absorption, because no model has a parameter to represent it. The MGG oracle h^2 (0.464) is lower than GM's (0.51) because Strehler-Mildvan compensation constrains the spread of intrinsic lifespans. The bias magnitude is statistically indistinguishable across model families (+9–10 pp), indicating that this is a general property of calibrate-then-extrapolate estimation under omitted extrinsic heterogeneity.

The bias is also robust to the extrinsic frailty construction itself. The default DGP uses multiplicative log-normal frailty; we tested two alternatives — additive perturbation ($c_i = m_{\text{ex}} + \sigma_\gamma \gamma_i$) and gamma-distributed frailty — holding mean extrinsic hazard fixed at m_{ex}. Across 10 replications per construction, the bias is $+8.1 \pm 0.4$ pp (log-normal), $+8.7 \pm 0.4$ pp (additive), and $+7.6 \pm 0.6$ pp (gamma), with all 95% CIs overlapping (Appendix, Section A.12). The 1 pp spread across constructions, compared to a 8 pp mean effect, confirms that the mechanism does not depend on the log-normal parameterization.

4.7 Anchored quantification

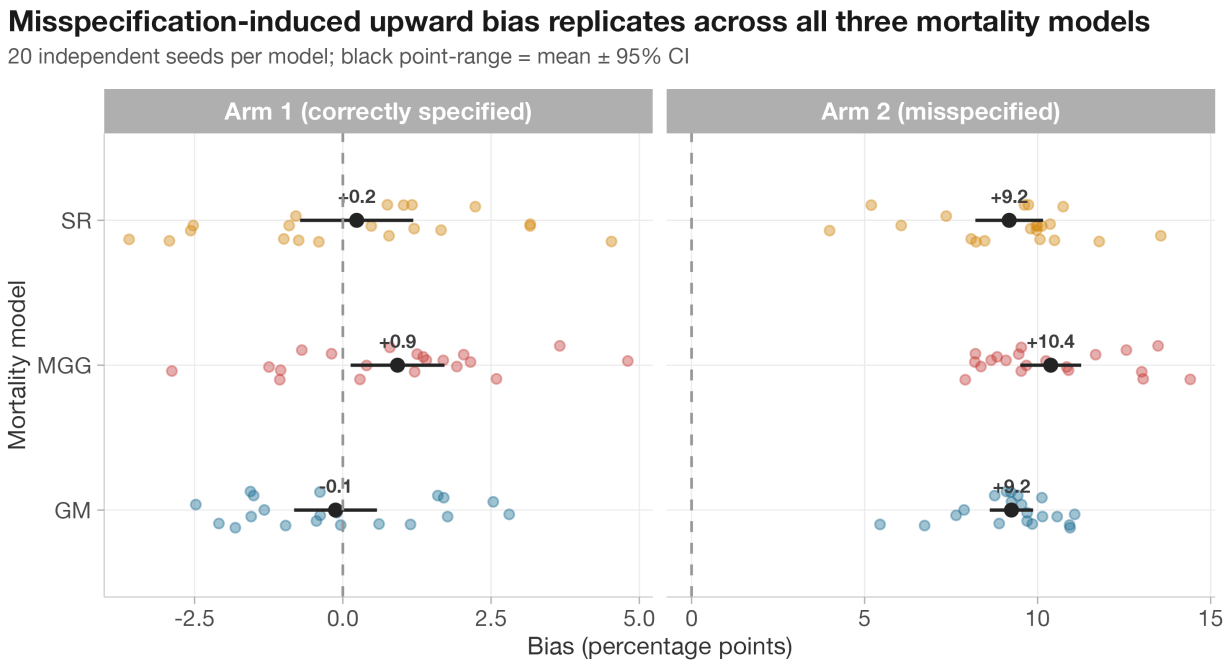
Within the externally informed restricted sensitivity regime ($\sigma_\gamma \in [0.30, 0.65]$, $\rho \in [0.20, 0.50]$), the bias ranges from +2.8 to +18.9 pp, with a mean of 9 pp (Figure 6). At the upper-bound point estimate ($\sigma_\gamma = 0.65$, $\rho = 0.35$), the bias is +15.7 pp. Across the full exploratory sweep, the bias reaches +18.4 pp at $\rho = 0.8$, $\sigma_\gamma = 0.55$ ($h^2 = 0.69$).

These results do not resolve the true intrinsic heritability, but they establish that Shenhar's estimate is biased upward by a quantitatively meaningful amount — sufficient to materially reduce their 50% estimate (by +15.7 pp at the upper-bound point estimate, and by 9 pp on average across the anchored regime). The analysis also yields three falsifiable predictions: (1) a two-component reanalysis that separately models extrinsic frailty should reduce intrinsic h^2 by a comparable amount; (2) the net bias should be larger in cohorts with higher infection-mortality share; and (3) if the genetic correlation $\rho < 0$, the net bias reverses sign (Corollary B3 in Section B). Figure 3 (panel B) confirms this last prediction: bias transitions smoothly from negative (downward) to positive (upward) as ρ crosses zero, with the sign reversal occurring near $\rho = 0$.

As a stress test, we extended the sensitivity sweep beyond the anchored ceiling to $\sigma_\gamma = 1.47$, the value implied by the tetrachoric bridge to liability-scale infection-mortality heritability (Figure 7, panel B; Appendix, Section A.11). Bias continues to increase monotonically, reaching $+32 \pm 0.5$ pp (106.7% inflation of σ_θ) at the bridge-implied value. While such extreme hazard heterogeneity may exceed what is biologically plausible under the stylized DGP, this confirms that the reported anchored range (+2.8 to +18.9 pp) is conservative: if the true extrinsic heterogeneity lies closer to the liability-scale implication, the bias is strictly larger, not smaller.

[3] We adopt the compensatory form $a = a_0^q$, consistent with Shenhar et al.'s preprint; see Appendix Section A.4 for implementation details.

(A)



(B)

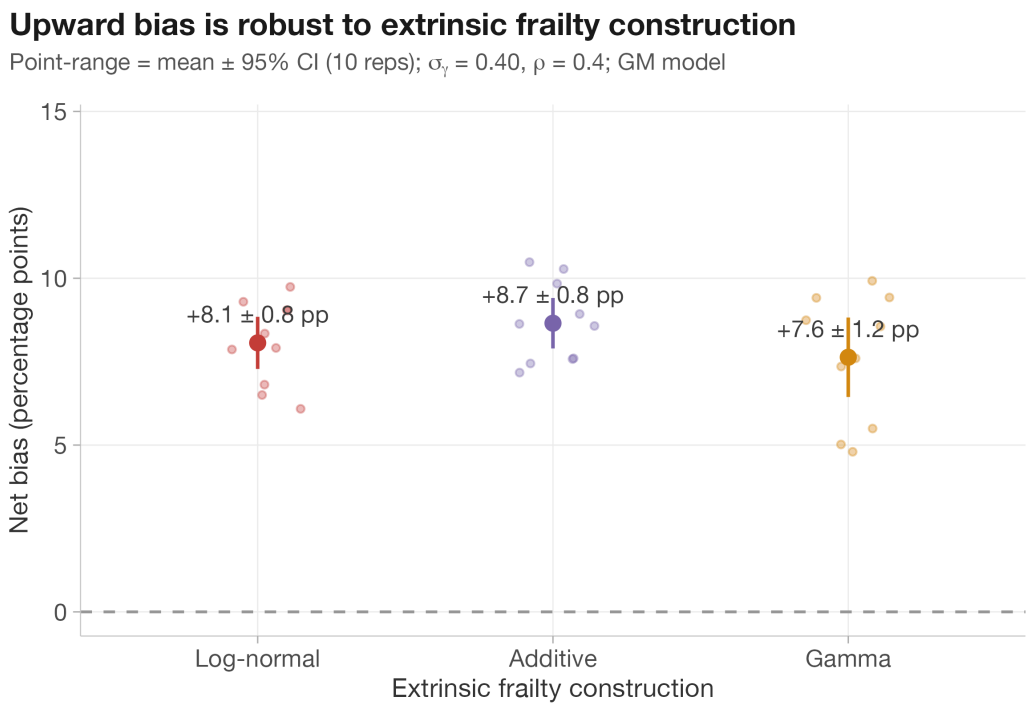


Figure 5: Robustness across model and distributional choices. **(A)** Misspecification-induced upward bias replicates across all three mortality models (20 independent seeds per model). Left: Arm 1 (correctly specified) clusters near zero. Right: Arm 2 (misspecified) is consistently +9–10 pp with tight 95% CIs entirely above +8 pp. Black point-range: mean ± 95% CI; colored dots: individual seeds. **(B)** Functional-form robustness (GM model, 10 replications per construction): log-normal, additive, and gamma extrinsic frailty constructions produce overlapping CIs, confirming that the bias does not depend on the parametric form of extrinsic heterogeneity.

Table 4: Model-generality validation. The misspecification bias (Arm 2 vs oracle) is positive for all three frameworks. All values are point estimates from a single canonical seed, retained for internal consistency with the decomposition and controls in §4.1–4.5. Replicated 20-seed means with 95% CIs (§4.6, Figure 5) are slightly higher (e.g., GM $+9.2 \pm 0.3$ pp vs +7.6 pp here); the difference reflects Monte Carlo variability across seeds, not a systematic discrepancy.

Model	Oracle h^2	Arm 1 h^2	Arm 1 bias	Arm 2 h^2	Arm 2 bias
Gompertz–Makeham	0.51	0.472	−3.7 pp	0.586	**+7.6 pp**
MGG (Shenhar)	0.464	0.459	−0.5 pp	0.556	**+9.2 pp**
SR (Shenhar)	0.577	0.569	−0.8 pp	0.655	**+7.8 pp**

5 Discussion

5.1 What is threatened in Shenhar's estimate (and what is not)

Shenhar's **mathematical framework** is valid: under its stated assumptions, setting $m_{\text{ex}} = 0$ genuinely reveals higher intrinsic heritability. The vulnerability is empirical. If extrinsic mortality susceptibility is heritable and pleiotropic with intrinsic aging — conditions supported by immunogenetic evidence — then the single-parameter calibration conflates two distinct genetic contributions, so the extrapolated "intrinsic" estimate reflects intrinsic aging variance plus whatever heritable extrinsic susceptibility was absorbed into the intrinsic parameter during calibration, rather than intrinsic aging alone.

The evidence shows that this is not merely a theoretical possibility but a quantitatively characterized mechanism with specific boundary conditions, dose-response behavior, and model-general expression.

An apparent objection arises from Shenhar et al.'s own supplementary analysis (their Fig S2), which shows that "variation in extrinsic mortality alone fails to account for the observed correlations" — varying m_{ex} as the sole source of individual heterogeneity cannot reproduce observed r_{MZ}. A reader might interpret this as evidence against heritable extrinsic susceptibility driving twin concordance. In fact, their finding is *consistent with* — and a prerequisite for — the bias mechanism we identify. Shenhar's Fig S2 tests whether extrinsic heterogeneity *alone* (without intrinsic heterogeneity) can produce the observed correlations; the answer is no. Our mechanism asks a different question: what happens when both θ and γ exist in the true DGP but the fitted model omits γ and estimates only σ_θ? Because extrinsic heterogeneity alone is insufficient to explain the correlations (as Shenhar confirm), the calibrated model has no channel through which to represent the extrinsic component of twin concordance — which is precisely why the one-parameter calibration absorbs it into σ_θ. Shenhar's finding establishes a necessary condition for the absorption mechanism: if γ variation *could* explain the correlations on its own, the model might not need to absorb it into the intrinsic parameter.

5.2 How large is the bias in plausible regimes?

Several bias magnitudes appear throughout this paper; they are not competing estimates but successive refinements at three levels of generality:

1. **Default demonstration** (+7.6 pp net). At the default parameters ($\sigma_\gamma = 0.40$, $\rho = 0.4$), the misspecification produces +11.4 pp of upward inflation offset by 3.7 pp of downward attenuation from extrinsic noise (§4.1). This is the worked example; it establishes the mechanism but makes no claim about the empirically correct parameter values.
2. **Anchored sensitivity regime** (+2.8 to +18.9 pp, mean 9 pp). Within the externally informed restricted range ($\sigma_\gamma \in [0.30, 0.65]$, $\rho \in [0.20, 0.50]$), the bias varies by roughly an order of magnitude

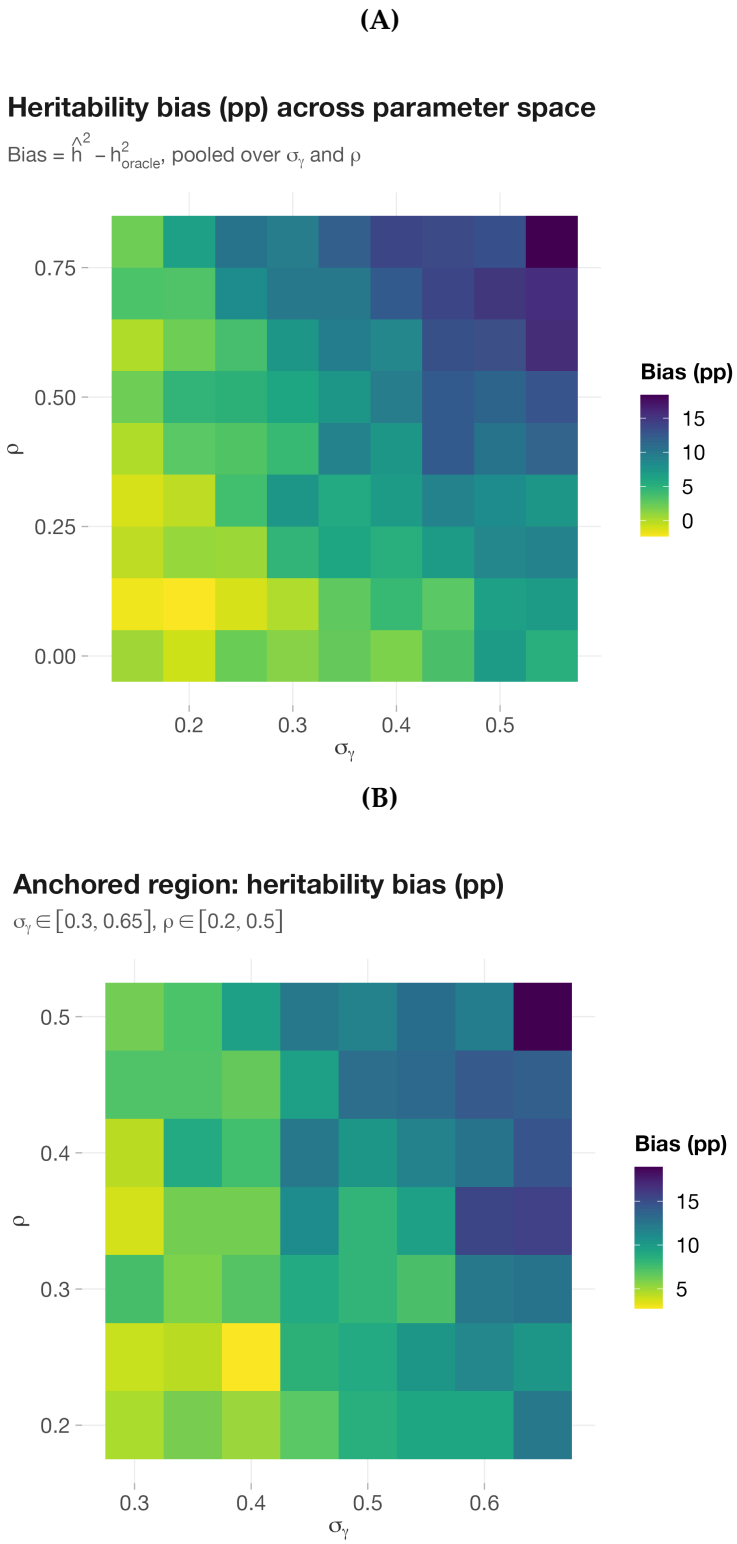


Figure 6: Sensitivity sweeps. **(A)** Full exploratory sweep: bias increases with both ρ and σ_γ, reaching +18.4 pp at $\rho = 0.8$, $\sigma_\gamma = 0.55$. **(B)** Anchored regime ($\sigma_\gamma \in [0.30, 0.65]$, $\rho \in [0.20, 0.50]$): bias ranges from +2.8 to +18.9 pp (mean 9 pp); point estimate ($\sigma_\gamma = 0.65$, $\rho = 0.35$): +15.7 pp.

(§4.7). At the upper-bound point estimate ($\sigma_\gamma = 0.65$, $\rho = 0.35$), it reaches +15.7 pp. This range is our primary quantitative result.

3. **Alternative calibration robustness** (still positive, magnitude changes). Multi-target calibration fitting five summary statistics yields +6.4 pp, versus +10 pp under r_{MZ}-only calibration on the

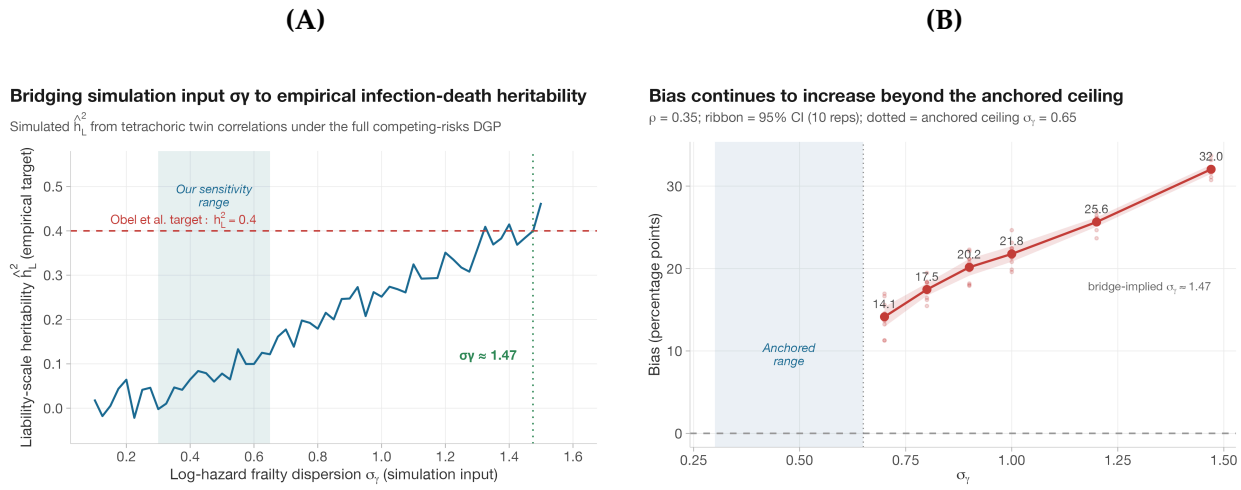


Figure 7: Conservative anchoring. **(A)** Bridging σ_γ to empirical infection-death heritability: matching Obel et al.'s $h^2_L \approx 0.40$ (dashed red) requires $\sigma_\gamma \approx 1.47$ (dotted green) — more than double the sensitivity ceiling of 0.65 (shaded band). **(B)** Stress test beyond the anchored ceiling ($\rho = 0.35$; 10 replications per level; ribbon = 95% CI). Bias continues to increase monotonically, reaching +32 pp at the bridge-implied σ_γ. Together, the panels confirm that the anchored range (+2.8 to +18.9 pp) is deliberately conservative within the explored framework.

same simulation draw (Section A.6). The small difference from the main analysis's +7.6 pp reflects Monte Carlo variability across independent draws, not a methodological discrepancy. The direction is robust; only the magnitude shifts with the fitting criterion.

Under even the most cautious assumptions within this range, the bias can materially reduce Shenhar's 50% estimate. Because the bridge analysis (Figure 7) confirms that published infection-mortality heritability implies extrinsic heterogeneity well above our sensitivity ceiling, the anchored regime is deliberately conservative within the explored framework: it was intentionally set well below the bridge-implied parameterization to ensure understated rather than overstated conclusions. An extended stress test confirms that bias continues to increase monotonically beyond $\sigma_\gamma = 0.65$, reaching +32 pp at the bridge-implied $\sigma_\gamma \approx 1.47$ (Figure 7, panel B).

Crucially, this upward bias is not an idiosyncrasy of any one mortality architecture. Under 20-seed replication, Arm 2 inflation is consistently +9–10 pp across all three model families (§4.6, Figure 5), and all three extrinsic frailty constructions produce overlapping CIs (+7.6 to +8.7 pp; Appendix, Section A.12) — indicating that the bias is a general property of calibrate-then-extrapolate estimation under omitted extrinsic heterogeneity.

5.3 What reanalysis would resolve it

All-cause twin survival data alone cannot uniquely identify a decomposition of genetic variance into "intrinsic" versus "extrinsic" pathways without additional structure — distributional signatures can reveal the presence of misspecification (§5.5), but not the partition itself. The most direct resolution would be a two-component reanalysis of Shenhar's cohort data — for example, a competing-risks model with an extrinsic frailty term shared within twin pairs, evaluated via posterior predictive checks on MZ/DZ bivariate survival surfaces. Incorporating cause-of-death information, polygenic scores for infection susceptibility, or hierarchical competing-risks likelihoods would further sharpen identification. The true intrinsic heritability could plausibly be lower than Shenhar's 50% if extrinsic susceptibility is heritable and unmodeled. Ruby et al.'s estimates (< 7%) highlight that multiple, distinct biases can operate in the same downward direction: they identify assortative mating as inflating *observed* twin correlations, while we identify omitted extrinsic heterogeneity as inflating the *calibrated* intrinsic parameter. These two corrections are analytically independent — ours addresses the calibrate-

then-extrapolate step, theirs addresses the input data — and are potentially additive. Applying our correction alone reduces Shenhar's 50% estimate by +2.8 to +18.9 pp depending on parameter assumptions; if assortative mating further inflates the twin correlations that Shenhar calibrates to, the combined effect could push the estimate substantially lower. However, Ruby et al. use genealogical (non-twin) data with different methodology, so direct combination requires care.

5.4 Low modern extrinsic mortality does not resolve the calibration bias

Modern extrinsic mortality rates ($m_{\text{ex}} \approx 3 \times 10^{-4}$; B. Shenhar, personal communication) are far lower than historical levels, and our dose-response analysis confirms that bias is negligible at such levels. However, this concerns the **evaluation point** ($m_{\text{ex}} \to 0$), not the **estimation point**. Shenhar's estimate is obtained by calibrating to **historical cohorts** where $m_{\text{ex}} \approx 0.004$, then extrapolating. The misspecification enters at calibration and is carried forward. The magnitude of Shenhar's own correction — from observed $h^2 \approx 0.20$ to intrinsic $h^2 \approx 0.50$ — demonstrates that extrinsic mortality is not negligible in the identifying cohorts.

5.5 Limitations

The calibration uses stochastic bisection on r_{MZ} rather than full likelihood. Joint $(r_{\text{MZ}}, r_{\text{DZ}})$ calibration yields a slightly larger bias (§3.3), and the structural argument (§2.2) predicts that any fitting criterion converges to an inflated pseudo-true σ_θ under positive pleiotropy. To verify this prediction directly, we re-estimated σ_θ using a multi-target minimum-distance calibration that simultaneously fits five summary statistics (r_{MZ}, mean and SD of age-at-death, q_{25}, q_{75}): the bias persists at +6.4 pp, versus +10 pp under r_{MZ}-only calibration on the same draw (Appendix, Section A.6; the small difference from Table 3's +7.6 pp reflects Monte Carlo variability across independent simulation draws). Moreover, the misspecified model's best-compromise fit leaves detectable residual misfit (e.g., q_{25} underpredicted by 2.2 years), confirming that the presence of omitted extrinsic frailty is in principle detectable from distributional features beyond twin correlations — though detecting misspecification is not the same as uniquely identifying the intrinsic/extrinsic decomposition, which requires the additional structure discussed in §5.3.

The anchoring of σ_γ and ρ to published liability-scale and GWAS estimands is approximate, and the sensitivity range ($\sigma_\gamma \in [0.30, 0.65]$, $\rho \in [0.20, 0.50]$) is restricted below empirically motivated levels rather than calibrated to them. We evaluate robustness by sweeping both parameters across wide ranges, and the bias is positive throughout the upper-left triangle of the parameter space (where pleiotropy and extrinsic heterogeneity are both non-negligible). COVID-19 severity provides an existence proof that genetic correlation between infection susceptibility and aging pathways is non-negligible, but it represents a single novel pathogen with specific immunological characteristics (ACE2-mediated entry, cytokine storm). Pre-antibiotic-era infection mortality in Danish cohorts born 1870–1900 was dominated by tuberculosis, pneumonia, diphtheria, and enteric infections — pathogens engaging different arms of the immune system. GWAS evidence strengthens the pleiotropic case: signals in the HLA class II region (HLA-DQA1/DRB1) reach genome-wide significance for both TB susceptibility (Schurz et al., 2024) and parental lifespan (Joshi et al., 2017), consistent with shared immunogenetic architecture, while Nordic-specific TB associations are documented in FinnGen (Tervi et al., 2023). The ρ range we adopt is informed by the convergence of COVID-19, HLA-longevity overlap, and Sørensen adoption-study evidence (Petersen et al., 2010; Sørensen et al., 1988), not by quantitative transfer from any single pathogen to historical cohorts.

Our simulation is framed around heritable extrinsic frailty — the case most directly motivated by immunogenetic evidence — but the calibration-absorption mechanism operates on any source of

within-pair concordance in mortality-relevant extrinsic susceptibility, whether genetic or shared-environmental (§4.3, footnote). The genetic case is our primary target; the broader familial generalization is a secondary implication that widens the scope of the vulnerability but does not change its structure.

The DGP assumes MZ twins share extrinsic susceptibility identically (correlation 1) and DZ twins at 0.5, paralleling the standard ACE model for intrinsic frailty. In reality, within-pair concordance in extrinsic susceptibility reflects both genetic sharing and individual-specific exposure noise (e.g., different infection contacts), so the effective within-pair correlation is below 1 even for MZ twins. Lower effective concordance would reduce the bias magnitude; Control A (independent γ) provides the zero-concordance boundary. The true effect lies between Control A and Arm 2; our estimates are therefore upper bounds conditional on the other DGP parameters.

The extrinsic frailty DGP uses multiplicative log-normal heterogeneity ($c_i = m_{\text{ex}} \cdot \exp(\sigma_\gamma \gamma_i - \sigma_\gamma^2/2)$). To verify that the bias is not driven by this distributional choice, we tested two alternative constructions: additive perturbation ($c_i = m_{\text{ex}} + \sigma_\gamma \gamma_i$, truncated at zero) and gamma-distributed frailty, each with mean extrinsic hazard held fixed at m_{ex}. Across 10 replications per construction, the bias is $+8.1 \pm 0.4$ pp (log-normal), $+8.7 \pm 0.4$ pp (additive), and $+7.6 \pm 0.6$ pp (gamma) — all with overlapping 95% CIs (Appendix, Section A.12). The 1 pp spread across constructions, relative to a 8 pp mean effect, confirms that the qualitative conclusion does not depend on the parametric form of extrinsic heterogeneity.

6 Conclusion

1. Shenhar's framework may be mathematically valid under its assumptions, but its "intrinsic" estimate is not empirically identified as purely intrinsic if heritable extrinsic frailty exists and is omitted at calibration. The vulnerability is empirical, not logical.
2. When extrinsic mortality susceptibility is heritable and pleiotropic with intrinsic aging, the single-parameter calibration absorbs both sources of mortality-relevant familial concordance, inflating the intrinsic parameter by 22.1%.
3. The bias decomposes into 3.7 pp of downward attenuation from extrinsic noise and +11.4 pp of upward inflation from heritable confounding, netting +7.6 pp.
4. Across three mortality model families and 20-seed Monte Carlo replication, Arm 1 bias is indistinguishable from zero while Arm 2 is consistently +9–10 pp (all 95% CIs $\{>\} + 8$ pp); the bias is also robust to alternative extrinsic frailty constructions (log-normal, additive, gamma).
5. The bias scales monotonically with the heritable fraction of extrinsic mortality, crossing zero near 15–20% heritable; the anchored sensitivity range (+2.8 to +18.9 pp) is conservative relative to bridge-implied levels.
6. The bias persists without survival conditioning (+3.1 pp at cutoff = 0), showing that concordant-survivor restriction is not necessary.
7. A correctly specified two-component model recovers σ_θ to within −3.2% of its true value, eliminating the inflation component while retaining the same attenuation seen under correct specification — confirming that the net upward bias is specifically from omitting extrinsic frailty.
8. Negative genetic correlation ($\rho < 0$) reverses the sign — a falsifiable prediction; sign reversal appears in all three model families.

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APPENDICES

A Appendix: Computational implementation

A.1 Pipeline architecture

All simulations use the `{targets}` reproducible pipeline framework (Landau 2021) with `{crew}` for parallel execution on 10 local workers (Apple M4 Pro, 14 cores, 24 GB RAM). The pipeline defines 194 targets organized in a directed acyclic graph (DAG), with static branching (`tar_map`) for the 81-cell sensitivity sweep and 56-cell anchored sweep. A single master seed (66829) propagates deterministic sub-seeds to every target, ensuring full reproducibility.

R 4.4.2 is used throughout. Key packages: `MASS` (multivariate normal twin parameters), `lamW` (Lambert W closed-form GM solver), `mvtnorm`, `Rcpp`.

A.2 Gompertz–Makeham solver

Lifespan inversion for the Gompertz–Makeham model uses the Lambert W function. The cumulative hazard $H_i(t) = c_i t + \left(\frac{a_i}{b}\right)\left(e^{bt} - 1\right) = y_i$ (where $y_i = -\log(V_i)$, $V_i \sim \text{Uniform}(0,1)$, $a_i = \exp(\theta_i)$, and c_i is the individual extrinsic hazard) has the closed-form solution:

$$T_i = \frac{y_i}{c_i} + \frac{a_i}{bc_i} - \frac{1}{b} W\left(\frac{a_i}{c_i} \cdot \exp\left(\frac{a_i}{c_i} + b\frac{y_i}{c_i}\right)\right) \tag{4}$$

For pure Gompertz ($c_i = 0$): $T_i = \left(\frac{1}{b}\right)\log\left(1 + b\frac{y_i}{a_i}\right)$. When $\log W_{\text{arg}} > 500$, the Lambert W function is computed in log space via Newton iteration ($w + \log w = z$) to avoid floating-point overflow. A bisection fallback handles remaining edge cases.

A.3 SR Euler–Maruyama solver (Rcpp)

The saturating-removal (SR) damage model requires numerical integration of the stochastic differential equation:

$$dx = \left(\eta t - \frac{\beta x}{x + \kappa}\right) dt + \sqrt{2\varepsilon} dW \tag{5}$$

with death occurring when damage x exceeds the individual threshold $X_{c,i}$ or an independent Poisson extrinsic event occurs. We integrate via Euler–Maruyama with timestep $\Delta t = 1/52$ (weekly).

The inner simulation loop is compiled to C++ via `Rcpp` for performance. The C++ implementation uses a Xoroshiro128+ pseudorandom number generator for the per-individual noise and extrinsic-event draws, avoiding the overhead of R's scalar RNG interface. A compact alive-index array skips deceased individuals at each timestep. This yields a measured ≈4.6× speedup over the equivalent vectorized R loop (0.54 s vs 2.5 s per 10,000-pair simulation). Each `{crew}` worker lazy-compiles the C++ source on first SR invocation; the compilation is cached for subsequent calls within the same worker process.

A.4 MGG solver and Strehler–Mildvan parameterization

The Makeham–Gamma–Gompertz model uses the closed-form cumulative hazard from Shenhar et al.'s Supplementary Materials. The Strehler–Mildvan coupling sets $b_i = q_i b_0$; the remaining question is how a_i depends on q_i. Shenhar et al.'s bioRxiv preprint (v1) writes the compensatory form in exponential notation: $a = a_0^q$. Taking logs gives the equivalent $\log a = q \cdot \log a_0$; in the published *Science* supplement, however, the q appears in the denominator — $\log a = (\log a_0)/q$ — yielding $a = a_0^{1/q}$. This is the kind of slip that can occur when converting between exponential and log-linear forms during manuscript preparation, and the preprint's form is clearly intended: only $a = a_0^q$ produces Strehler–Mildvan compensation (higher q raises b but *lowers* a, because $a_0 = 10^{-5} < 1$), yields hazard-curve crossing, and reproduces Shenhar et al.'s reported twin correlations and heritability estimates.

Under $a = a_0^q$, higher q raises b but lowers a, producing the characteristic hazard-curve crossing at $t^* = -\frac{\ln(a_0)}{b_0} \approx 100$ years — the hallmark of Strehler–Mildvan compensation. Figure 8 illustrates the consequences. Panel A confirms that $a = a_0^q$ produces converging hazard curves while $a = a_0^{1/q}$ produces diverging ones. Panel B shows cohort survival: the alternative form creates a bimodal mortality pattern in which high-q individuals die prematurely (no compensation to restrain their hazard). Panel C confirms the identification: with identical dispersion ($\sigma_q = 0.27$) and no extrinsic mortality, $a = a_0^q$ yields $r_{\mathrm{MZ}} = 0.43$ and Falconer $h^2 = 0.47$ — consistent with Shenhar et al.'s published twin correlations — whereas $a = a_0^{1/q}$ yields $r_{\mathrm{MZ}} = 0.92$ and $h^2 = 1.01$, exceeding unity.

We adopt $a = a_0^q$ throughout. The inverse CDF is solved via hybrid bisection–Newton iteration with 40 steps.

A.5 Calibration

All calibration uses 40-iteration bisection on the respective dispersion parameter (σ_θ for GM, σ_q for MGG, σ_{X_c} for SR) to match a target r_{MZ}. The stochastic bisection uses common random numbers (CRN): the same RNG seed is used for each candidate parameter value within a bisection run, reducing variance in the comparison.

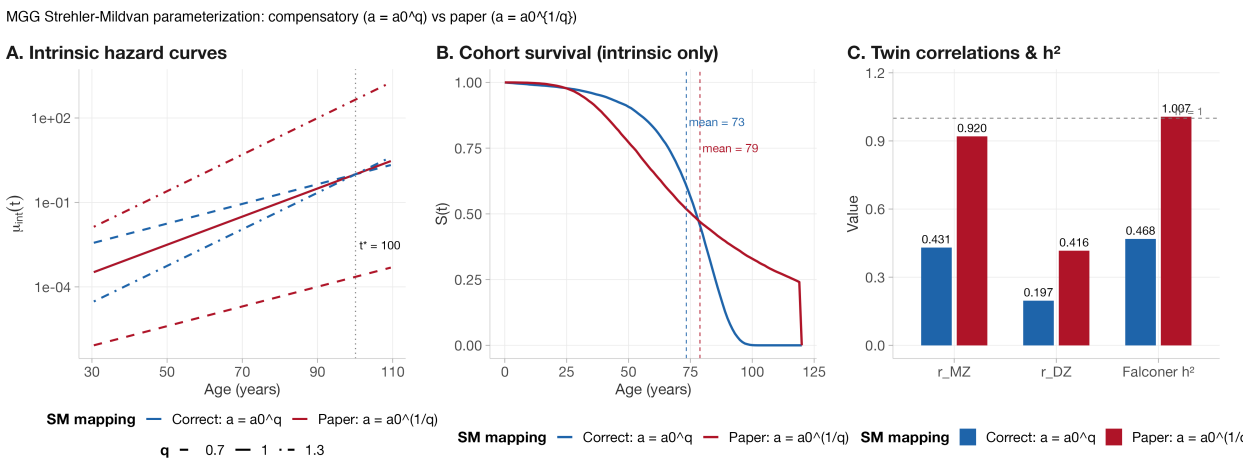

Figure 8: Strehler–Mildvan parameterization comparison. (A) Intrinsic hazard curves for $q \in \{0.7, 1.0, 1.3\}$: the compensatory form $a = a_0^q$ (blue) produces convergence at t^*; the literal supplement formula $a = a_0^{1/q}$ (red) produces divergence. (B) Cohort survival ($n = 20000$, $m_{\mathrm{ex}} = 0$): the literal formula creates early mortality in high-q individuals. (C) Oracle twin correlations and Falconer h^2 ($\sigma_q = 0.27$, $m_{\mathrm{ex}} = 0$): $a = a_0^q$ yields $h^2 = 0.47$, consistent with Shenhar et al.'s published values; $a = a_0^{1/q}$ yields $h^2 = 1.01$.

Table 5: Multi-target vs r_{MZ}-only calibration under Arm 2 DGP. Both methods produce upward-biased $\hat{h}^2$; multi-target calibration attenuates but does not eliminate the net upward bias. The r_{MZ}-only baseline here uses a separate simulation draw from the main analysis (Table 3) to ensure an internally consistent comparison; the 2 pp difference in bias (+10 vs +7.6 pp) reflects Monte Carlo variability across draws. Oracle $h^2 \approx 0.51$.

Calibration criterion	σ_θ fit	Inflation	$\hat{h}^2$ at $m_{\mathrm{ex}} = 0$	Bias (pp)
Multi-target (5 statistics)	1.47	+16.8%	0.573	**+6.4 pp**
r_{MZ}-only	1.582	+25.7%	0.609	**+10 pp**

A.6 Multi-target calibration robustness

A potential concern is that the upward bias in Arm 2 could be an artifact of calibrating the misspecified intrinsic-only model by matching only r_{MZ}. To test this, we re-estimated σ_θ using a multi-target calibration that simultaneously fits five summary statistics computed from concordant-survivor MZ pairs: r_{MZ}, the mean and standard deviation of age-at-death, and the 25th and 75th percentiles. The misspecified model's sole free parameter σ_θ was chosen by one-dimensional optimization (Brent method) minimizing a weighted sum of squared errors across all five targets, with common random numbers ensuring a deterministic objective and fair comparison to the r_{MZ}-only calibration.

Because the misspecified model has only one degree of freedom and five fitting targets, the system is overidentified — the optimizer finds a best-compromise σ_θ that cannot match all targets simultaneously.

Under multi-target calibration, the inferred intrinsic-only heritability remained biased upward ($\hat{h}^2 = 0.573$, +6.4 pp relative to the oracle $h^2 \approx 0.51$), although less than under r_{MZ}-only calibration ($\hat{h}^2 = 0.609$, +10 pp). The best-compromise fit leaves detectable residual misfit: the 25th percentile of age-at-death is underpredicted by 2.2 years, and the goodness-of-fit RMSE across all targets is 1.13 years. These residuals confirm that the omitted extrinsic frailty is in principle identifiable from distributional features, and that the upward heritability bias is not specific to the choice of r_{MZ} as the sole calibration target.

A.7 Grid specifications

Sensitivity sweeps use the following grids:

All models use 50,000 pairs per zygosity. The SR Euler–Maruyama solver is $O(n \cdot T/\Delta t)$; with the Rcpp implementation, each 50,000-pair SR simulation completes in ≈2.5 s.

A.8 Hardware and runtime

Simulations were run on an Apple M4 Pro (14 cores, 24 GB unified memory) under macOS 26 (Tahoe). The `{crew}` controller dispatches up to 10 parallel workers. The pipeline defines 194 targets with static branching for sweeps and per-model MC targets dispatched as independent crew jobs. Inner parallelism (`mclapply` with $\min(n_{\mathrm{seeds}}, 4)$ cores) is used within long-running MC targets to exploit the remaining cores on each worker. The full pipeline — including all SR sweeps, 20-seed multi-model MC replication, and robustness analyses — completes in approximately 2 hours.

A.9 Heritable-fraction replication

The heritable-fraction sensitivity analysis (§4.5, Figure 4) was replicated with 5 independent seeds per fraction level (21 levels from 0 to 1 by 0.05). Each replication uses an independent RNG seed derived from the master seed with a per-replication offset, ensuring distinct twin draws. The 95% CIs shown

Table 6: Grid specifications for all parameter sweeps and robustness checks.

Grid	Specification
Full sweep	$\rho \in \{0, 0.1, ..., 0.8\} \times \sigma_\gamma \in \{0.15, 0.20, ..., 0.55\}$ (81 cells)
Anchored sweep	$\rho \in \{0.20, 0.25, ..., 0.50\} \times \sigma_\gamma \in \{0.30, 0.35, ..., 0.65\}$ (56 cells)
Dose–response	15 m_{ex} values from 0 to 0.012
Negative ρ	$\rho \in \{-0.60, -0.55, ..., 0.80\}$ by 0.05 (29 points)
m_{ex} split	21 heritable fractions from 0 to 1 by 0.05
σ_γ bridge	57 values from 0.1 to 1.5 by 0.025
Extended σ_γ	7 values from 0.70 to 1.47 ($\rho = 0.35$; 10 reps each)
Functional form	3 constructions (log-normal, additive, gamma; 10 reps each)
Multi-model MC	20 seeds × 3 models (GM, MGG, SR)
Heritable fraction (hi-res)	21 levels from 0 to 1 by 0.05 (5 reps each)

in Figure 4 are t-based intervals ($\overline{x} \pm t_{n-1,0.025} \cdot \text{SE}$) computed across the 5 replicates at each fraction level; with $n = 5$ the t-multiplier is 2.78 rather than the normal-approximation 1.96. This replication confirmed that the "dip" near 50% heritable visible in the original single-seed analysis was a Monte Carlo artifact; with replication, bias increases monotonically.

A.10 Multi-model Monte Carlo uncertainty

To quantify Monte Carlo uncertainty across mortality model families, we ran 20 independent seed replications for each of the three models (GM, MGG, SR). Each seed defines a complete Arm 1/Arm 2 pipeline: fresh twin-pair generation, calibration (40-iteration bisection with CRN), and evaluation at $m_{\text{ex}} = 0$. Oracle h^2 is computed from a separate pure-intrinsic simulation. The three per-model targets (`mc_unc_gm`, `mc_unc_mgg`, `mc_unc_sr`) are dispatched as independent `{crew}` jobs for full parallelism. Within each target, the 20-seed loop uses `mclapply` with 4 cores to further exploit available hardware. Reported CIs are normal-approximation intervals ($\overline{x} \pm 1.96 \cdot \text{SE}$) across the 20 seeds; with $n = 20$ the normal and t-based intervals are nearly identical. For the SR model, the Rcpp Euler–Maruyama solver is used throughout, with each seed receiving a distinct fast-seed derived from the R RNG state.

A.11 Extended σ_γ stress test

The extended σ_γ sweep (§4.7, Figure 7 (panel B)) evaluates bias at 7 grid points from $\sigma_\gamma = 0.70$ to $\sigma_\gamma = 1.47$ (the bridge-implied value), with ρ fixed at 0.35 and 10 replications per grid point. This complements the main anchored sweep (which caps at $\sigma_\gamma = 0.65$) by characterizing the bias trajectory beyond the intentionally conservative sensitivity ceiling. Each replication uses an independent seed; CIs are normal-approximation intervals across the 10 replicates.

A.12 Functional form variants

The functional-form robustness check (§4.6, §5.5) tests three constructions for extrinsic frailty heterogeneity, each with mean extrinsic hazard held fixed at $m_{\text{ex}} = 0.004$:

1. **Log-normal** (baseline): $c_i = m_{\text{ex}} \cdot \exp\left(\sigma_\gamma \gamma_i - \sigma_\gamma^2/2\right)$, with $\gamma_i \sim \mathcal{N}(0, 1)$. This is the default DGP used throughout the paper.

2. **Additive**: $c_i = m_{\text{ex}} + \sigma_\gamma \gamma_i$, truncated at zero. The additive construction models heterogeneity as a shift in the extrinsic hazard rather than a multiplicative perturbation.
3. **Gamma**: $c_i \sim \text{Gamma}(k, k/m_{\text{ex}})$, where k is chosen to match the coefficient of variation of the log-normal construction at the same σ_γ. This ensures that the first two moments of the extrinsic hazard distribution are matched across constructions.

Each construction is evaluated with 10 independent replications at the default parameters ($\sigma_\gamma = 0.40$, $\rho = 0.4$) under the GM mortality model. The 95% CIs are normal-approximation intervals across replicates. All three constructions use the same calibration machinery (40-iteration bisection on r_{MZ}) and the same evaluation protocol (Falconer h^2 at $m_{\text{ex}} = 0$).

B Appendix: Formal proof of calibration inflation

This appendix provides a first-order (delta-method) proof that omitting heritable extrinsic frailty from the fitted model inflates the calibrated intrinsic dispersion parameter. The qualitative predictions are corroborated by exact simulation throughout the main text.

B.1 Setup: true DGP and fitted model

True data-generating process. Let $b > 0$ and $m_{\text{ex}} > 0$ be fixed. For individual i, define intrinsic frailty $\theta_i = \sigma_\theta(\alpha_i + \beta_i)$, where $\alpha_i \sim \mathcal{N}(0,1)$ is the additive genetic component and $\beta_i \sim \mathcal{N}(0,1)$ is the non-shared environmental component. For an MZ pair $(1,2)$: $\alpha_1 = \alpha_2$ and $\beta_1 \perp \beta_2$. Note $\text{Var}(\theta_i) = 2\sigma_\theta^2$; under this parameterization, σ_θ is the per-component scale factor, not the standard deviation of θ_i — note $\text{SD}(\theta_i) = \sqrt{2}\sigma_\theta$. This convention separates the genetic (α) and environmental (β) components with symmetric unit-variance scaling.

Define extrinsic frailty $\gamma_i = \gamma_i^{\text{gen}} + \gamma_i^{\text{env}}$ with γ^{gen} shared identically in MZ pairs, $\gamma_1^{\text{env}} \perp \gamma_2^{\text{env}}$, and $\text{Var}(\gamma^{\text{gen}}) = \text{Var}(\gamma^{\text{env}}) = 1/2$, so $\text{Var}(\gamma) = 1$. Assume $\text{Corr}(\alpha, \gamma^{\text{gen}}) = \rho \in [-1, 1]$, with all other cross-components independent. The individual extrinsic hazard is $c_i := m_{\text{ex}} \exp\big(\sigma_\gamma \gamma_i\big)$.

True hazard:

$$h_i^{\text{true}}(t) = \exp(\theta_i)\exp(bt) + c_i \tag{6}$$

Fitted (misspecified) model. Shenhar's fitted class uses $h_i^{\text{fit}}(t; \hat{\sigma}_\theta) = \exp\big(\tilde{\theta}_i\big)\exp(bt) + c$ with $\tilde{\theta}_i = \hat{\sigma}_\theta(\alpha_i + \beta_i)$ and constant c (no individual extrinsic variation). We take c to be fixed at the population mean, matching the procedure of calibrating σ_θ while holding b and c fixed. The calibration adjusts $\hat{\sigma}_\theta$ to match the observed MZ twin correlation r_{MZ} from the true DGP, then extrapolates by setting $c = 0$.

B.2 Linearization via delta method

Let $L = \log T$. The cumulative hazard under the true DGP is:

$$\Lambda_i(t) = \frac{\exp(\theta_i)}{b}(\exp(bt) - 1) + c_i \cdot t \tag{7}$$

Death occurs at T_i where $\Lambda_i(T_i) = U_i$, $U_i \sim \text{Exp}(1)$. Define:

$$F(T; \theta, \gamma, U) = \frac{\exp(\theta)}{b}(\exp(bT) - 1) + m_{\text{ex}} \exp\big(\sigma_\gamma \gamma\big) \cdot T - U = 0 \tag{8}$$

Since $\partial F / \partial T = h^{\text{true}}(T) > 0$, the implicit function theorem guarantees $T = T(\theta, \gamma, U)$ is differentiable. Differentiating with respect to a generic parameter $\varphi \in \{\theta, \gamma\}$:

$$\frac{\partial T}{\partial \varphi} = -\frac{(\partial \Lambda / \partial \varphi)}{h(T)} \tag{9}$$

Intrinsic slope. With $\partial \Lambda / \partial \theta = \Lambda_{\text{int}}(T)$, define:

$$\kappa_\theta := \Lambda_{\text{int}} \frac{T}{T \cdot h_{\text{total}}(T)} = -\frac{\partial \log T}{\partial \theta} \tag{10}$$

Extrinsic slope. By the chain rule ($\partial \Lambda / \partial \gamma = \sigma_\gamma \cdot \Lambda_{\text{ext}}(T)$), define κ_γ as the base extrinsic slope *without* the σ_γ factor:

$$\kappa_\gamma := \Lambda_{\text{ext}} \frac{T}{T \cdot h_{\text{total}}(T)} = \frac{c_i}{h_{\text{total}}(T)} \tag{11}$$

so that $-\partial \log T / \partial \gamma = \sigma_\gamma \kappa_\gamma$.

Slope approximation. Under the true model, $h^T_{\text{total}}(T) = \exp(\theta)\exp(bT) + c_i$; under the fitted model, $h^F_{\text{total}}(T) = \exp\left(\tilde{\theta}\right)\exp(bT) + c$. Define $r_\kappa = \kappa^F_\theta / \kappa^T_\theta$; note $r_\kappa \to 1$ as $m_{\text{ex}}/[\exp(\theta)\exp(bT)] \to 0$. The analysis treats this difference as higher-order relative to the omitted-variable term in the empirically relevant regime.

First-order expansion. True DGP:

$$L_i \approx \overline{L}^T - \kappa_\theta \sigma_\theta (\alpha_i + \beta_i) - \kappa_\gamma \sigma_\gamma \gamma_i + \varepsilon^T_i \tag{12}$$

Fitted model:

$$\tilde{L}_i \approx \overline{L}^F - \kappa_\theta \hat{\sigma}_\theta (\alpha_i + \beta_i) + \varepsilon^F_i \tag{13}$$

where ε^T, ε^F capture higher-order terms and residual stochasticity from U.

B.3 Variance-covariance decomposition for MZ pairs

Covariance normalization. Given $\alpha \sim \mathcal{N}(0,1)$ and $\gamma^{\text{gen}} \sim \mathcal{N}(0, 1/2)$ with $\text{Corr}(\alpha, \gamma^{\text{gen}}) = \rho$:

$$\text{Cov}(\alpha, \gamma^{\text{gen}}) = \rho \cdot 1 \cdot \frac{1}{\sqrt{2}} = \frac{\rho}{\sqrt{2}} \tag{14}$$

For an MZ pair, the shared components are α (identical) and γ^{gen} (identical); the non-shared components are $\beta_1 \perp \beta_2$ and $\gamma^{\text{env}}_1 \perp \gamma^{\text{env}}_2$.

True MZ covariance (C_{true}). Only shared components contribute:

$$C_{\text{true}} = \kappa^2_\theta \sigma^2_\theta \underbrace{\text{Var}(\alpha)}_{=1} + \kappa^2_\gamma \sigma^2_\gamma \underbrace{\text{Var}(\gamma^{\text{gen}})}_{=1/2} + 2\kappa_\theta \kappa_\gamma \sigma_\theta \sigma_\gamma \underbrace{\text{Cov}(\alpha, \gamma^{\text{gen}})}_{=\rho/\sqrt{2}} \tag{15}$$

$$C_{\text{true}} = \kappa^2_\theta \sigma^2_\theta + \frac{1}{2}\kappa^2_\gamma \sigma^2_\gamma + \sqrt{2}\rho \kappa_\theta \kappa_\gamma \sigma_\theta \sigma_\gamma \tag{16}$$

True DGP variance (V_{true}). All components contribute. Both α and β contribute to intrinsic variance (each with $\text{Var} = 1$, independent), giving a factor of 2:

$$V_{\text{true}} = 2\kappa^2_\theta \sigma^2_\theta + \kappa^2_\gamma \sigma^2_\gamma + \sqrt{2}\rho \kappa_\theta \kappa_\gamma \sigma_\theta \sigma_\gamma + \sigma^2_{\varepsilon,T} \tag{17}$$

The cross-term $\sqrt{2}\rho \kappa_\theta \kappa_\gamma \sigma_\theta \sigma_\gamma$ appears once (not doubled) because only α (not β) is correlated with γ^{gen}, and γ^{env} is independent of everything.

Fitted model. With only $\tilde{\theta} = \hat{\sigma}_\theta(\alpha + \beta)$ and no extrinsic variation:

$$C_{\text{fit}} = \kappa^2_\theta \hat{\sigma}^2_\theta, \quad V_{\text{fit}} = 2\kappa^2_\theta \hat{\sigma}^2_\theta + \sigma^2_{\varepsilon,F} \tag{18}$$

Key structural asymmetry. C_{true} contains extra terms from shared extrinsic frailty (γ^{gen}) and pleiotropy (ρ) that C_{fit} lacks; V_{true} contains extra non-shared extrinsic variance that V_{fit} lacks. The net effect on $r_{\text{MZ}} = C/V$ depends on relative magnitudes.

B.4 Calibration equation and inflation condition

Lemma (existence and uniqueness). In the fitted model, $r^{\text{fit}}_{\text{MZ}}(\hat{\sigma}_\theta)$ is strictly increasing in $\hat{\sigma}_\theta$ and takes values in $(0, 1/2)$ for $\sigma^2_{\varepsilon,F} > 0$. Hence a unique finite $\hat{\sigma}_\theta$ exists if and only if $r^{\text{true}}_{\text{MZ}} < 1/2$, equivalently $V_{\text{true}} - 2C_{\text{true}} > 0$. We assume this throughout.

Correlation matching. The calibration sets $r^{\text{fit}}_{\text{MZ}}(\hat{\sigma}_\theta) = r^{\text{true}}_{\text{MZ}}$:

$$\frac{\kappa_\theta^2 \hat{\sigma}_\theta^2}{2\kappa_\theta^2 \hat{\sigma}_\theta^2 + \sigma_{\varepsilon,F}^2} = \frac{C_{\text{true}}}{V_{\text{true}}} \tag{19}$$

Let $\hat{V} = \kappa_\theta^2 \hat{\sigma}_\theta^2$ and $\Omega_F = \sigma_{\varepsilon,F}^2$. Cross-multiplying (using $V_{\text{true}} - 2C_{\text{true}} > 0$):

$$\hat{V} = \kappa_\theta^2 \hat{\sigma}_\theta^2 = \frac{C_{\text{true}} \cdot \sigma_{\varepsilon,F}^2}{V_{\text{true}} - 2C_{\text{true}}} \tag{20}$$

Define shorthands: $V_\theta^T = \kappa_\theta^2 \sigma_\theta^2$ (intrinsic), $V_\gamma^T = 1/2\kappa_\gamma^2 \sigma_\gamma^2$ (extrinsic), $C_\rho = \sqrt{2}\rho\kappa_\theta\kappa_\gamma\sigma_\theta\sigma_\gamma$ (pleiotropy). Then:

$$C_{\text{true}} = V_\theta^T + V_\gamma^T + C_\rho, \quad V_{\text{true}} - 2C_{\text{true}} = \sigma_{\varepsilon,T}^2 - C_\rho \tag{21}$$

The feasibility condition requires $\sigma_{\varepsilon,T}^2 > C_\rho$. Even at the upper bounds of the anchored regime ($\rho = 0.50$, $\kappa_\theta \approx 0.8$, $\kappa_\gamma \approx 0.05$, $\sigma_\theta \approx 0.27$, $\sigma_\gamma = 0.65$), $C_\rho \approx \sqrt{2} \cdot 0.50 \cdot 0.80 \cdot 0.05 \cdot 0.27 \cdot 0.65 \approx 0.005$, which is one to two orders of magnitude below typical residual log-lifespan variance; the condition is comfortably satisfied throughout the anchored parameter space. Substituting:

$$\kappa_\theta^2 \hat{\sigma}_\theta^2 = \frac{\left(V_\theta^T + V_\gamma^T + C_\rho\right) \cdot \sigma_{\varepsilon,F}^2}{\sigma_{\varepsilon,T}^2 - C_\rho} \tag{22}$$

Inflation condition. Inflation ($\hat{\sigma}_\theta > \sigma_\theta$) requires $\hat{V} > V_\theta^T$. When $\sigma_{\varepsilon,T}^2 \approx \sigma_{\varepsilon,F}^2 =: \sigma_\varepsilon^2$ (valid when extrinsic hazard is small relative to intrinsic at typical death ages), this simplifies to:

$$\left(V_\gamma^T + C_\rho\right) \cdot \sigma_\varepsilon^2 > -V_\theta^T \cdot C_\rho \tag{23}$$

When $\rho \geq 0$: $C_\rho \geq 0$ and $V_\gamma^T > 0$, so the LHS is strictly positive and the RHS is ≤ 0. The inequality holds.

Proposition B1 (local inflation under first-order approximation). *Under the first-order linearization of log-lifespan and when slope and residual-noise differences between the true and fitted models are second-order, omission of heritable extrinsic frailty ($\sigma_\gamma > 0$) with non-negative pleiotropy ($\rho \geq 0$) inflates the calibrated intrinsic dispersion: $\hat{\sigma}_\theta > \sigma_\theta$.*

Scope of the equal-noise approximation. The approximation $\sigma_{\varepsilon,T}^2 \approx \sigma_{\varepsilon,F}^2$ becomes quantitatively unreliable when extrinsic frailty variance is large — for example, at $\sigma_\gamma \approx 1.47$ (the bridge analysis estimate that matches published infection-mortality heritability). In such regimes, the analytical bound isolates the direction and mechanism of bias, but the exact simulation results — not the first-order approximation — constitute the operative quantitative statement on bias magnitude.

For the case $\rho < 0$, see Corollary B3 below.

B.5 Remark on concordant survivor truncation

Let $A = \{T_1 > t_0, T_2 > t_0\}$ denote concordant survival to age t_0. Truncation selects against high-frailty pairs. Because shared components (α, γ^{gen}) are carried by both twins jointly while non-shared components (β, γ^{env}) are selected independently for each twin, truncation tends to reduce non-shared variance relatively more than shared variance. Empirically, the sign of the bias is preserved and often amplified. Since the bias mechanism operates at the calibration step regardless of sample restriction, truncation is not required for the main result.

B.6 Monotonicity of extrapolated $h^2(\sigma)$

Proposition B2. *Under the first-order approximation with fixed local slope κ and frailty-independent residual variance σ_ε^2, the extrapolated Falconer heritability $h^2(\sigma) = 2(r_{\text{MZ}} - r_{\text{DZ}})$ is strictly increasing in σ for $\sigma > 0$ at the extrapolation point ($c = 0$, pure Gompertz).*

Proof. Under pure Gompertz with $c = 0$, MZ twins share α (correlation 1) and DZ twins share α with correlation 0.5:

$$r_{\mathrm{MZ}}(\sigma) = \frac{\kappa^2\sigma^2}{2\kappa^2\sigma^2 + \sigma_\varepsilon^2}, \quad r_{\mathrm{DZ}}(\sigma) = \frac{0.5 \cdot \kappa^2\sigma^2}{2\kappa^2\sigma^2 + \sigma_\varepsilon^2} \tag{24}$$

$$h^2(\sigma) = 2(r_{\mathrm{MZ}} - r_{\mathrm{DZ}}) = \frac{\kappa^2\sigma^2}{2\kappa^2\sigma^2 + \sigma_\varepsilon^2} \tag{25}$$

$$\frac{dh^2}{d\sigma} = \frac{2\kappa^2\sigma \cdot \sigma_\varepsilon^2}{(2\kappa^2\sigma^2 + \sigma_\varepsilon^2)^2} > 0 \quad \text{for all } \sigma > 0 \tag{26}$$

If calibration inflates $\hat{\sigma}_\theta > \sigma_\theta$ (Proposition B1), it inflates the extrapolated heritability estimate within the same first-order framework. □

B.7 Sign reversal under negative pleiotropy

Corollary B3. *If $\rho < 0$ and sufficiently negative, the omitted extrinsic component reduces the observed MZ correlation, causing the fitted model to underestimate intrinsic dispersion: $\hat{\sigma}_\theta < \sigma_\theta$.*

Under the equal-residual-noise simplification ($\sigma_{\varepsilon,T}^2 \approx \sigma_{\varepsilon,F}^2 = \sigma_\varepsilon^2$), the exact inflation boundary is:

$$\rho > -\frac{\kappa_\gamma\sigma_\gamma}{2\sqrt{2}\kappa_\theta\sigma_\theta} \cdot \frac{\sigma_\varepsilon^2}{\kappa_\theta^2\sigma_\theta^2 + \sigma_\varepsilon^2} \tag{27}$$

Downward bias occurs when ρ falls below this threshold. A sufficient condition for sign reversal (stronger but easier to interpret) that does not depend on σ_ε^2 is $V_\gamma^T + C_\rho < 0$:

$$\rho < -\frac{\kappa_\gamma\sigma_\gamma}{2\sqrt{2}\kappa_\theta\sigma_\theta} \tag{28}$$

This corresponds to the high-noise limit $\sigma_\varepsilon^2 \gg \kappa_\theta^2\sigma_\theta^2$, where the residual-dependent factor approaches 1.

B.8 Interpretation under empirically motivated parameter regimes

The theoretical analysis is a first-order (delta-method) result. Its qualitative predictions are confirmed by exact simulation across the biologically anchored parameter space. The manuscript's sensitivity regime uses $\sigma_\gamma \in [0.30, 0.65]$, $\rho \in [0.20, 0.50]$, and $m_{\mathrm{ex}} \in [0.001, 0.012]$. Since $\rho > 0$ throughout, Proposition B1 predicts upward bias in $\hat{\sigma}_\theta$ — confirmed quantitatively at 5–14 pp across all three model families. The sign-reversal threshold (Corollary B3) requires negative ρ, outside the empirically motivated range. The first-order analysis provides the direction and mechanism; the simulation provides exact magnitudes beyond the linear approximation.

B.9 Summary of formal results

Proposition B1 (local inflation). Omission of heritable extrinsic frailty ($\sigma_\gamma > 0$) with non-negative pleiotropy ($\rho \geq 0$) inflates the calibrated intrinsic dispersion under first-order linearization.

Proposition B2 (monotonicity). The extrapolated Falconer $h^2(\sigma)$ is strictly increasing in σ. Combined with B1, this implies upward bias in the extrapolated intrinsic heritability.

Remark (truncation). Concordant survivor truncation is not required for the bias but empirically amplifies it.

Corollary B3 (sign reversal). Sufficiently negative pleiotropy reverses the bias direction.

Simulations verify these qualitative predictions beyond the local approximation, across all three model families and throughout the biologically anchored parameter space.